REAL ANALYSIS II · SEMESTER 5 · MODULE 2

Prepare: Differentiation

Everything you need before you open the question paper, from Plus Two calculus and Real Analysis I up to Bartle & Sherbert §6.1 (6.1.1–6.1.7) and §6.2 (6.2.1–6.2.13): the derivative, Rolle, the Mean Value Theorem and Darboux.

File 1 of 3	**Prepare** (this file): learn the ideas
File 2	Model paper: related and new questions
File 3	Answer key: full solutions

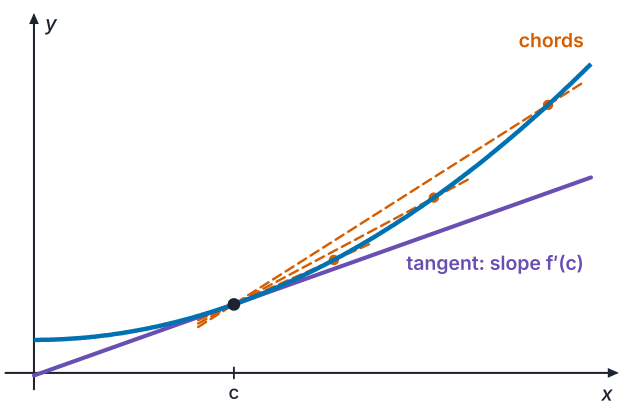
Each answer in File 3 starts with topic codes such as **T12** **P04** **M06** . If a code is new to you, read that topic here first.

One idea for the whole module

THE DERIVATIVE IS A SLOPE THAT IS THE LIMIT OF CHORD SLOPES

Take a point c on a graph. Join it to a nearby point x by a straight line (a **chord**). Its slope is the **difference quotient** $\frac{f(x)-f(c)}{x-c}$. Slide x towards c. If these slopes settle down to one number, that number is the **derivative** $f'(c)$.

Said another way: near c, the graph is **best approximated by a straight line** of slope $f'(c)$, and the error shrinks faster than the distance $|x-c|$ does.



The derivative. The chord from (c, f(c)) to (x, f(x)) has slope (f(x) − f(c))/(x − c). As x → c the chords turn into the tangent; its slope is f'(c) = $\lim_{x \to c}$ (f(x) − f(c))/(x − c).

THE CHAIN THAT CARRIES THE MODULE

Everything after the definition hangs on one chain:

interior maximum ⇒ slope 0 (T10) ⇒ **Rolle** (T11) ⇒ **Mean Value Theorem** (T12) ⇒ constant, increasing, extrema, inequalities, roots, Darboux (T13–T24).

The Mean Value Theorem says that a smooth arc from A to B has a point where the tangent is parallel to the chord AB. Whenever you know something about f' **everywhere** on an interval, the MVT turns it into something about f.

The chain in one table

> Read down: each topic is proved from the one above it.

STEP	WHAT IT SAYS	BUILT FROM
T01 Definition	$f'(c) = \lim_{x \to c} \dfrac{f(x) - f(c)}{x - c}$	limits (P04)
T02 Differentiable ⇒ continuous	a jump is impossible; the converse is false ($\lvert x \rvert$)	T01
T05 Carathéodory	$f(x) - f(c) = \varphi(x)(x - c)$, with φ continuous at c	T01
T06 Chain rule	$(g \circ f)'(c) = g'(f(c))\, f'(c)$	T05
T10 Interior extremum	peak or valley inside ⇒ $f' = 0$	T01, local

		sign
T11 Rolle	$f(a) = f(b) \Rightarrow$ some $f'(c) = 0$	T10, Max–Min
T12 MVT	$f(b) - f(a) = f'(c)(b - a)$	T11
T13 – T15	$f' = 0 \Rightarrow$ constant; $f' \geq 0 \Leftrightarrow$ increasing; sign change of $f' \Rightarrow$ extremum	T12
T16 – T21	roots, estimates, inequalities	T12
T22 – T24	Darboux: derivatives have the intermediate value property	T10, T22

Codes and map

P01–P13 Plus Two and Real Analysis I bridge

M01–M10 Maths tools

T01–T24 Module 2 topics

C5 a counterexample from the library

Q12 question 12 of the model paper

SYLLABUS UNIT	BARTLE	TOPICS
7. The derivative	6.1.1–6.1.7	T01–T09
8. Extrema, Rolle, MVT	6.2.1–6.2.6	T10–T13
9. Monotone, tests, applications	6.2.7–6.2.9	T14–T17
10. Inequalities, Darboux	6.2.10–6.2.13	T18–T24
11. Selected problems	all of the above	whole file

T01 Derivative at a point (6.1.1)

T02 Differentiable ⇒ continuous (6.1.2)

T03 Sum, product, quotient rules (6.1.3)

T04 Power rule, finite sums and products (6.1.4)

T05 Carathéodory's characterisation (6.1.5)

T06 Chain rule (6.1.6)

T07 Derivative of $|f|$, sgn and absolute values (6.1.7)

T08 Derivatives of tan, sec, cot, csc (6.1.7)

T09 $x^2 \sin(1/x)$ (6.1.7)

T10 Relative extremum, Interior Extremum Theorem (6.2.1–6.2.2)

T11 Rolle's Theorem (6.2.3)

T12 Mean Value Theorem (6.2.4)

T13 $f' = 0$ ⇒ constant (6.2.5–6.2.6)

T14 Monotonicity criterion (6.2.7)

T15 First Derivative Test (6.2.8)

T16 Rolle and the roots of f (6.2.9a)

T17 MVT for estimates (6.2.9b)

T18 Inequalities: e^x and $\ln x$ (6.2.10a)

T19 Inequalities: $\sin x$, Lipschitz (6.2.10b)

T20 Bernoulli's inequality (6.2.10c)

T21 Weighted AM–GM (6.2.10d)

T22 Local sign lemma (6.2.11)

T23 Darboux's Theorem (6.2.12)

T24 A function that is not a derivative (6.2.13)

At least 15 of the 70 external marks. The six old papers contain no direct Module 2 question: this module was taught in Semester 2 calculus before. The model paper therefore uses the 2024 Calculus paper as "related" and Bartle's starred exercises.

The whole module in one story

THE IDEA

1. The derivative at c is the limit of chord slopes $\frac{f(x)-f(c)}{x-c}$; where it exists, f is continuous, and the usual rules (sum, product, quotient, chain) follow from limit laws and Carathéodory's trick (T01–T06).

2. Some functions are differentiable but badly behaved: $|x|$ has a corner, and $x^2\sin(1/x)$ is differentiable everywhere with a derivative that is not continuous (T07–T09).

3. At a peak or valley inside an interval the slope is zero (T10); with the Max–Min Theorem this gives Rolle (T11) and then the Mean Value Theorem (T12).

4. From the MVT: zero derivative means constant, positive derivative means increasing, and a sign change of f' marks an extremum (T13–T15); also roots, estimates and inequalities (T16–T21).

5. Derivatives cannot jump: Darboux says f' takes every value between two of its values, so signum is not a derivative of anything (T22–T24).

The hypotheses to check every time

FOR ROLLE (T11) AND THE MVT (T12)

- ☐ f is **continuous on the closed interval** $[a, b]$
  a jump at an end breaks it C18
- ☐ f is **differentiable on the open interval** (a, b)
  $|x|$ on $[-1, 1]$ has a corner C18
- ☐ for Rolle only: $f(a) = f(b)$
  $f(x) = x$ on $[0, 1]$ has slope 1 everywhere C18
- ☐ for the Interior Extremum Theorem: the point is **interior**
  x on $[0, 1]$ has extrema at the ends, slope 1 N7

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Tap a line to jump there. Every topic heading has an **index ↑** link back here, and every topic chip such as T5 in the text is a link too.

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P01 The derivative in Plus Two

index ↑

Slope of the tangent, speed, and the difference quotient.

REMIND ME · PLUS TWO CH 5

- **Slope** of a line through two points = rise ÷ run $= \dfrac{y_2 - y_1}{x_2 - x_1}$.
- For a function f and a point c, the line joining $(c, f(c))$ to $(x, f(x))$ has slope $\dfrac{f(x) - f(c)}{x - c}$. This is the **difference quotient** (the "chord slope").
- The **derivative** $f'(c)$ is the number these slopes approach as x comes close to c. It is the slope of the tangent, and the instantaneous rate of change.
- Plus Two writes $\dfrac{f(c+h) - f(c)}{h}$ with $h \to 0$. Bartle writes $x \to c$. They are the same thing with $h = x - c$.
- **Tangent line** at c: $y - f(c) = f'(c)\,(x - c)$ (a line through $(c, f(c))$ with slope $f'(c)$).

CHECK YOURSELF

For $f(x) = x^2$ and $c = 3$, simplify the chord slope and find the tangent line.

$\dfrac{x^2 - 9}{x - 3} = x + 3 \to 6$. So $f'(3) = 6$ and the tangent is $y - 9 = 6(x - 3)$.

P02 Plus Two rules you may quote

index ↑

> Bartle proves the rules; the standard derivatives are assumed known.

REMIND ME · PLUS TWO CH 5

Df or f' both mean "the derivative of f". For functions f, g and a constant α:

- Sum, constant multiple: $(f+g)' = f' + g'$, $(\alpha f)' = \alpha f'$.
- Product: $(fg)' = f'g + fg'$. Quotient (where $g \neq 0$): $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$.
- Chain: $(g \circ f)'(x) = g'(f(x))\, f'(x)$.

Standard derivatives (quote them without proof):

x^n, x^r	nx^{n-1}, rx^{r-1} (for $x > 0$ when r is not a whole number)
$\sin x$, $\cos x$	$\cos x$, $-\sin x$
e^x, $\ln x$	e^x, $1/x$ (for $x > 0$)

EXAM TRAP

Bartle says $\sin' = \cos$ and the others are "proved later" (6.1.7(d), Section 8.3). In an answer write: "by the standard derivatives of Plus Two". Do not try to prove them. Do not use a rule to prove the same rule (e.g. the product rule in the proof of the product rule).

CHECK YOURSELF

Differentiate $x^2 \sin x$.

Product rule: $2x \sin x + x^2 \cos x$.

P03 Graph shapes to recognise

index ↑

> You will be asked to picture these, so know what each looks like.

REMIND ME · PLUS ONE AND PLUS TWO

x^2, x^3	x^2 is a cup through the origin, flat at 0. x^3 is an S through the origin, flat at 0 but always climbing.
$\lvert x\rvert$	A V with a sharp corner at 0. Slope -1 on the left, $+1$ on the right.
$\operatorname{sgn} x$	-1 for $x < 0$, 0 at 0, $+1$ for $x > 0$. A jump at 0.
$\sqrt{x}$, $x^{1/3}$	$\sqrt{x}$ leaves 0 straight up (vertical tangent), only for $x \geq 0$. $x^{1/3}$ is an S-curve, also vertical at 0, defined for all x.
$1/x$	Two branches, undefined at 0, no point where the graph crosses the y-axis.
$\sin x$, $\cos x$	Waves between -1 and 1 with period 2π.
e^x, $\ln x$	e^x is always positive and rising. $\ln x$ is its mirror image in the line $y = x$.
$\sin(1/x)$	Wiggles infinitely fast near 0, always between -1 and 1.
$x^2 \sin(1/x)$	The same wiggle squeezed between the parabolas $y = \pm x^2$. Very flat at 0.

A HABIT WORTH HAVING

Before you compute, sketch. A corner means "no derivative here". A vertical tangent means "slope infinite here". A smooth wiggle means "derivative exists, but check it".

P04 Limits of functions (Real Analysis I)

index ↑

> The derivative is a limit, so we need these results in exact form.

REMIND ME · BARTLE 4.1, 4.2

- A point c is a **cluster point** of a set A if every interval round c contains points of A other than c.
- **Definition 4.1.4.** Let $A \subseteq \mathbb{R}$ and let c be a cluster point of A. For a function $f : A \to \mathbb{R}$, a real number L is said to be a **limit** of f at c if, given any $\varepsilon > 0$ there exists $\delta(\varepsilon) > 0$ such that if $x \in A$ and $0 < |x - c| < \delta(\varepsilon)$, then $|f(x) - L| < \varepsilon$.
- **Limit laws** (4.2.4): limits of sums, products and quotients (non-zero limit in the denominator) are the sum, product and quotient of the limits, *when each limit exists*.
- **Order** (4.2.6): if $f(x) \geq 0$ near c (for $x \neq c$) and $\lim_{x\to c} f$ exists, then $\lim_{x\to c} f \geq 0$. A strict "$>$" can become "$\geq$" in the limit.
- **Local sign** (4.2.9): if $\lim_{x\to c} f = L > 0$, there is a neighbourhood of c on which $f(x) > 0$ for $x \neq c$ (and similarly for $L < 0$).
- **Sequential criterion** (4.1.8): $\lim_{x\to c} f = L$ iff $f(x_n) \to L$ for **every** sequence $x_n \to c$ with $x_n \neq c$.
- **Squeeze**: $g \leq f \leq h$ near c and $\lim g = \lim h = L \Rightarrow \lim f = L$.

WHY WE NEED LOCAL SIGN AND ORDER

Local sign ("limit positive ⇒ function positive near c") is how a derivative's sign is turned into a statement about values (T10, T22). Order ("f ≥ 0 ⇒ limit ≥ 0") is how f increasing gives $f' \geq 0$ (T14).

P05 Sequences and the useful ones

index ↑

> Two sequences give a quick proof that a limit does not exist.

REMIND ME · BARTLE 3.1

- $x_n \to c$ means: for every $\varepsilon > 0$ there is K with $|x_n - c| < \varepsilon$ for all $n \geq K$.
- We always take $x_n \neq c$ when we use the sequential criterion (P04).

Sequences you will use again and again:

$x_n = \dfrac{1}{n}$	tends to 0
$x_n = \dfrac{1}{n\pi}$	tends to 0; $\sin \dfrac{1}{x_n} = \sin n\pi = 0$
$x_n = \dfrac{1}{2n\pi}$	tends to 0; $\cos \dfrac{1}{x_n} = \cos 2n\pi = 1$
$x_n = \dfrac{1}{(2n+1)\pi}$	tends to 0; $\cos \dfrac{1}{x_n} = \cos(2n+1)\pi = -1$
$x_n = c + \dfrac{1}{n}$	tends to c from the right, $x_n \neq c$

CHECK YOURSELF

What is $\lim n\{f(c+\frac{1}{n})-f(c)\}$ for $f(x)=x^2$, $c=1$?

$n\{(1+\frac{1}{n})^2-1\}=n\{\frac{2}{n}+\frac{1}{n^2}\}=2+\frac{1}{n}\to 2=f'(1).$

P06 Continuity (Module 1)

index ↑

Differentiability sits on top of continuity.

DEFINITION 5.1.1 · SAY IT WORD FOR WORD

Let $A\subseteq\mathbb{R}$, let $f:A\to\mathbb{R}$, and let $c\in A$. We say that f is **continuous at c** if, given any number $\varepsilon>0$ there exists $\delta>0$ such that if x is any point of A satisfying $|x-c|<\delta$, then $|f(x)-f(c)|<\varepsilon$.

In words: whatever band of half-height ε you are handed round f(c), there is a window of half-width δ round c whose points all land in the band. Equivalent: $\lim_{x\to c} f(x)=f(c)$.

REMIND ME · MODULE 1

- Sums, products, constant multiples and quotients (denominator $\neq 0$) of continuous functions are continuous.
- **Composition** (5.2.6): if f is continuous at c and g is continuous at $f(c)$, then $g\circ f$ is continuous at c.
- Polynomials, $\sin$, $\cos$, e^x, $|x|$ are continuous everywhere.
- Details: Module 1 prepare file, topics T01–T07.

EXAM TRAP

"Continuous" is a statement about *values*, "differentiable" about *slopes*. Every differentiable function is continuous (T02), but not the other way: $|x|$ at 0.

P07 The big theorems of Module 1

index ↑

> Rolle and Darboux are built on these. Know the statements.

MAXIMUM–MINIMUM THEOREM 5.3.4

Let $I = [a, b]$ be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Then f has an absolute maximum and an absolute minimum on I.

So there is a point $x^* \in I$ with $f(x) \leq f(x^*)$ for every $x \in I$. This is what makes Rolle and Darboux work.

BOLZANO'S INTERMEDIATE VALUE THEOREM 5.3.7

Let I be an interval and let $f : I \to \mathbb{R}$ be continuous on I. If $a, b \in I$ and if $k \in \mathbb{R}$ satisfies $f(a) < k < f(b)$, then there exists a point $c \in I$ between a and b such that $f(c) = k$.

REMIND ME · MODULE 1

- **Location of Roots** (5.3.5): f continuous on $[a, b]$ with $f(a) < 0 < f(b)$ (or the reverse) ⇒ a root in (a, b).
- **Lipschitz** (Definition 5.4.4): there is $K > 0$ with $|f(x) - f(u)| \leq K|x - u|$ for all $x, u \in A$. (T19 shows that a bounded derivative gives this.)
- Details: Module 1 prepare file, topics T08–T22.

WHAT 5.3.4 NEEDS

- ☐ f is **continuous**
  a jump can skip the maximum
- ☐ the interval is **closed**
  x on $(0, 1)$ has no maximum
- ☐ and **bounded**
  x on $[0, \infty)$ has no maximum

P08 Intervals, neighbourhoods and interior points

index ↑

> The words "interior" and "relative" are everywhere in §6.2.

REMIND ME · BARTLE 2.2, 2.5

- $V_\delta(c) = (c-\delta, c+\delta)$ is the **δ-neighbourhood** of c: all x with $|x-c| < \delta$.
- A point c of an interval I is **interior** if some neighbourhood $V_\delta(c)$ lies inside I. So an interior point is any point of I except an **endpoint**.
- $[a,b]$: interior is (a,b). (a,b): every point is interior. $[0,\infty)$: interior is $(0,\infty)$.
- "**Relative**" (or "local") means: compared only with nearby points, inside some neighbourhood.

WHY INTERIOR MATTERS

The derivative needs x to approach c from **both** sides. At an endpoint only one side exists, so a peak there need not have slope 0: $f(x) = x$ on $[0,1]$ peaks at 1 with slope 1 N7.

CHECK YOURSELF

Is 0 an interior point of $[0,1]$? Of $(-1,1)$?

No: every neighbourhood of 0 contains negative numbers, which are not in $[0,1]$. Yes for $(-1,1)$ (take $\delta = 1$).

P09 Inequalities and absolute value

index ↑

The rules you use on every inequality line. More practice: Module 1 inequality gym.

REMIND ME · PLUS ONE, BARTLE 2.2

- $|ab| = |a||b|$. $|a| < \varepsilon \iff -\varepsilon < a < \varepsilon$.
- **Triangle inequality**: $|a + b| \leq |a| + |b|$.
- **Multiplying**: by a positive number keeps the direction ($2 < 3 \Rightarrow 10 < 15$). By a negative number it **flips** ($2 < 3 \Rightarrow -4 > -6$).
- **Signs of products**: positive × positive = positive; negative × negative = positive; mixed = negative.
- **Strict and non-strict**: "$<$" says "strictly less". If one step in a chain is strict, the whole chain is strict.

MATHS TOOL · THE SIGN OF A QUOTIENT

If $q > 0$ then $(x - c)\,q$ has the same sign as $x - c$. So $x > c$ gives a positive product and $x < c$ a negative one. We use this in T10 and T22.

CHECK YOURSELF

If $f(x) - f(c) = (x - c) \cdot q$ with $q > 0$ and $x < c$, is $f(x) > f(c)$ or $f(x) < f(c)$?

$x - c < 0$ and $q > 0$, so the product is negative: $f(x) < f(c)$.

P10 Piecewise functions and the signum

index ↑

> Differentiability at a join is the commonest short question.

REMIND ME · PLUS ONE CH 2

- A **piecewise** function uses different formulas on different parts of the domain, such as $f(x) = x^2$ for $x \le 1$ and $2x - 1$ for $x > 1$.
- The point where the formulas change is the **join**. The value there is given by the piece whose condition includes the join (here $f(1) = 1$).
- **Signum**: $\operatorname{sgn} x = 1$ for $x > 0$, 0 for $x = 0$, -1 for $x < 0$. Then $|x| = x \operatorname{sgn} x$.

CHECK YOURSELF

For the f above, are the two pieces equal at 1? Do they have the same slope there?

Values: $1^2 = 1$ and $2(1) - 1 = 1$, so yes. Slopes: $2x = 2$ and 2, so yes. The join is smooth: f is differentiable at 1 with $f'(1) = 2$.

P11 Trigonometry values you need

index ↑

Angles in radians. Used in the $x^2 \sin(1/x)$ examples.

REMIND ME · PLUS TWO CH 2

- For every real x: $|\sin x| \le 1$ and $|\cos x| \le 1$.
- For each whole number n: $\sin n\pi = 0$, $\cos 2n\pi = 1$, $\cos(2n+1)\pi = -1$.
- $\sin \frac{\pi}{2} = 1$, $\cos \frac{\pi}{2} = 0$, $\cos 0 = 1$, $\sin 0 = 0$.
- $\tan x = \frac{\sin x}{\cos x}$, $\sec x = \frac{1}{\cos x}$, $\cot x = \frac{\cos x}{\sin x}$, $\csc x = \frac{1}{\sin x}$. $\sin^2 x + \cos^2 x = 1$.

CHECK YOURSELF

Evaluate $\cos \frac{1}{x}$ at $x = \frac{1}{3\pi}$ and at $x = \frac{1}{4\pi}$.

$\frac{1}{x} = 3\pi$: $\cos 3\pi = -1$. $\frac{1}{x} = 4\pi$: $\cos 4\pi = 1$.

P12 Exponential and logarithm

index ↑

Used in the MVT inequalities of T18.

REMIND ME · PLUS TWO CH 1, 5

- $e^0 = 1$, $e \approx 2.718$. $e^x > 0$ for every x. $e^x > 1$ for $x > 0$ and $e^x < 1$ for $x < 0$. e^x is increasing.
- $\ln x$ is defined only for $x > 0$. $\ln 1 = 0$, and $\ln x$ is increasing. $\ln x$ and e^x undo each other.
- $\frac{d}{dx} e^x = e^x$ and $\frac{d}{dx} \ln x = \frac{1}{x}$.

CHECK YOURSELF

Is $e^c > 1$ or < 1 for c between -1 and 0?

$c < 0$, so $e^c < 1$ (and $e^c > e^{-1} > 0$).

P13 Quadratics and sign charts

index ↑

Needed to decide where f' is positive or negative.

REMIND ME · CLASS 10, PLUS ONE

- $ax^2 + bx + c = 0$ has roots $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The number $D = b^2 - 4ac$ is the **discriminant**.
- If $D < 0$ there are no real roots, so the quadratic never crosses zero and keeps the sign of a for all x.
- **Sign chart**: mark the roots of each factor on a number line, then test one point in each interval. A factor changes sign only at its own root.

CHECK YOURSELF

What is the sign of $x(x-1)(x-2)$ on $(1, 2)$? And is $4x^2 - 15x + 18$ ever zero?

Test $x = 1.5$: $1.5 \times 0.5 \times (-0.5) < 0$, so negative. $D = 225 - 288 = -63 < 0$ and $a = 4 > 0$: always positive, never zero.

M01 Limit of a difference quotient

index ↑

Three steps, in this order.

MATHS TOOL

1. Write the quotient $\dfrac{f(x)-f(c)}{x-c}$ with the actual formulas.
2. **Simplify until the factor $x-c$ cancels** top and bottom. (You may cancel it because $x \neq c$ in a limit.)
3. Put $x=c$ in what is left.

Micro-example. $f(x)=x^2$: $\dfrac{x^2-c^2}{x-c} = \dfrac{(x-c)(x+c)}{x-c} = x+c \to 2c$.

Micro-example. $f(x)=3x$: $\dfrac{3x-3c}{x-c} = 3$ for every $x \neq c$, so the limit is 3.

EXAM TRAP

Do not put $x=c$ at the start: you get $\dfrac{0}{0}$. The whole point is to cancel the zero factor first.

CHECK YOURSELF

Find $f'(-1)$ for $f(x)=x^2$ by this method.

Quotient $= x+c$ with $c=-1$: tends to -2.

M02 Factorising and rationalising

index ↑

> How to make the factor $x - c$ appear.

MATHS TOOL

1. **Powers**: $x^n - c^n = (x-c)(x^{n-1} + x^{n-2}c + \cdots + c^{n-1})$. For $n = 3$: $x^3 - c^3 = (x-c)(x^2 + xc + c^2)$.
2. **Square roots**: multiply top and bottom by the "conjugate": $\sqrt{x} - \sqrt{c} = \dfrac{(\sqrt{x} - \sqrt{c})(\sqrt{x} + \sqrt{c})}{\sqrt{x} + \sqrt{c}} = \dfrac{x - c}{\sqrt{x} + \sqrt{c}}$.
3. **Fractions**: put over a common denominator: $\dfrac{1}{x} - \dfrac{1}{c} = \dfrac{c - x}{xc} = -\dfrac{x - c}{xc}$.

Micro-example: $\dfrac{\sqrt{x} - \sqrt{c}}{x - c} = \dfrac{1}{\sqrt{x} + \sqrt{c}} \to \dfrac{1}{2\sqrt{c}}$ $(c > 0)$.

Micro-example: $\dfrac{1/x - 1/c}{x - c} = \dfrac{-1}{xc} \to -\dfrac{1}{c^2}$.

CHECK YOURSELF

Find $\lim_{x \to c} \dfrac{x^{-2} - c^{-2}}{x - c}$ **for** $c \neq 0$.

$\dfrac{1}{x^2} - \dfrac{1}{c^2} = \dfrac{c^2 - x^2}{x^2 c^2} = \dfrac{-(x-c)(x+c)}{x^2 c^2}$. Divide by $x - c$: $\dfrac{-(x+c)}{x^2 c^2} \to \dfrac{-2c}{c^4} = -\dfrac{2}{c^3}$.

M03 One-sided limits and one-sided derivatives

index ↑

> At a corner or a join, compute both sides.

MATHS TOOL

- $x \to c^+$ means x approaches c from the right ($x > c$). $x \to c^-$ means from the left ($x < c$).
- A limit exists iff the left limit and the right limit exist and are **equal**.
- So $f'(c)$ exists iff the right quotient limit and the left quotient limit both exist and are equal.

Micro-example. $f(x) = |x|$ at 0: $\dfrac{|x| - 0}{x - 0} = \dfrac{|x|}{x}$, which is $+1$ for $x > 0$ and -1 for $x < 0$. The one-sided limits are $+1$ and -1: no derivative.

AT A JOIN OF A PIECEWISE FUNCTION, CHECK IN ORDER

☐ the two pieces have the same value at the join (continuity)

x^2 then $4\sqrt{x}$ at 1: values 1 and 4, stop, not differentiable **N3**

☐ the two one-sided derivatives agree

$2x + |x|$ at 0: slopes 3 and 1 **N3**

CHECK YOURSELF

Is $h(x) = 2x + |x|$ differentiable at 0?

$h = 3x$ for $x \geq 0$ and x for $x \leq 0$. Right slope 3, left slope 1: not equal. No.

M04 Limit laws and the squeeze

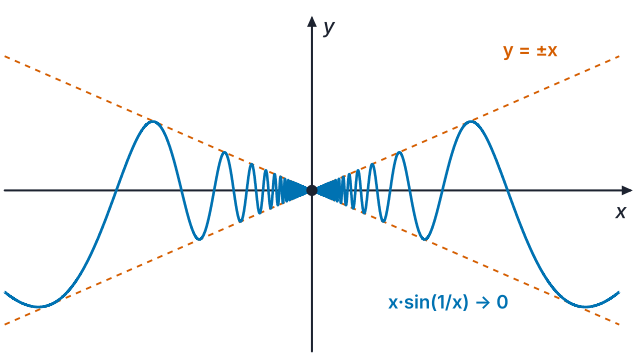
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> When a factor has no limit, use the squeeze, not the product law.

MATHS TOOL

- **Product law**: $\lim(fg) = \lim f \cdot \lim g$ *only when both limits exist.*
- **Bounded times zero is zero**: if $|g(x)| \le B$ and $f(x) \to 0$, then $f(x)g(x) \to 0$. Write $0 \le |fg| \le B|f| \to 0$ and squeeze.

Micro-example. $\sin\dfrac{1}{x}$ has no limit at 0, so we cannot split $x\sin\dfrac{1}{x}$. But $\left|x\sin\dfrac{1}{x}\right| \le |x| \to 0$, so $x\sin\dfrac{1}{x} \to 0$.



x·sin(1/x) is squeezed between −|x| and |x|, so it tends to 0 even though sin(1/x) has no limit. This is the squeeze step used to show that f′(0) = 0 in the x² sin(1/x) example.

CHECK YOURSELF

Find $\lim_{x \to 0} x^2 \cos \dfrac{1}{x}$.

$\left| x^2 \cos \dfrac{1}{x} \right| \le x^2 \to 0$, so the limit is 0.

M05 Sign analysis of the derivative

index ↑

> The routine behind "find the intervals of increase" and "find the extrema".

MATHS TOOL · FOUR STEPS

1. Find f' and factorise it.
2. Solve $f' = 0$ (and note where f' does not exist). Mark these points on a number line.
3. Test one point in each interval and record the sign of f'.
4. Read off: $f' > 0$ means rising, $f' < 0$ means falling. Include the end points of each interval (Bartle's "increasing" is non-strict).

Micro-example. $f(x) = x^3 - 3x$: $f' = 3(x - 1)(x + 1)$, zeros ± 1. Test: $x = -2$ gives $+9$; $x = 0$ gives -3; $x = 2$ gives $+9$. So increasing on $(-\infty, -1]$, decreasing on $[-1, 1]$, increasing on $[1, \infty)$.

WHY THIS IS ALLOWED

"Sign of f' gives rising or falling" is proved in T14, and "sign change gives extremum" in T15. For now, learn the routine; those topics supply the reasons.

CHECK YOURSELF

Where is $f(x) = x^2 - 3x + 5$ decreasing?

$f' = 2x - 3$, zero at $\frac{3}{2}$; $f' < 0$ for $x < \frac{3}{2}$. Decreasing on $(-\infty, \frac{3}{2}]$.

M06 An inequality from the MVT

index ↑

Five steps. This is the template for T17–T21 (compare with the gym, P09).

MATHS TOOL

The MVT (T12) says: for some c strictly between a and b, $f(b) - f(a) = f'(c)(b - a)$. We do not know c, but we do know where it lies.

1. Write $f(b) - f(a) = f'(c)(b - a)$ for the interval you want, with $a < c < b$.
2. **Bound** $f'(c)$ using $a < c < b$ (the derivative is monotone there, or $|\cdot| \leq 1$).
3. Multiply by $b - a > 0$, so the direction is kept.
4. The result is strict if the bound on $f'(c)$ is strict.
5. Treat $x = a$ (equality) and the other side of a (a different interval) separately.

Micro-example. For $0 \leq x, y \leq M$: $x^2 - y^2 = 2c\,(x - y)$ for some c between y and x, and $0 \leq 2c \leq 2M$, so $|x^2 - y^2| \leq 2M|x - y|$. (A Lipschitz bound.)

EXAM TRAP

The MVT gives *some* c, not the midpoint and not a number you can name. An inequality needs a bound on f' over the **whole** open interval N12.

CHECK YOURSELF

For $f(x) = x^3$, show $|x^3 - y^3| \leq 3|x - y|$ for $x, y \in [0, 1]$.

$x^3 - y^3 = 3c^2(x - y)$ with $c \in [0, 1]$, so $3c^2 \leq 3$. Hence $|x^3 - y^3| \leq 3|x - y|$.

M07 No limit, by two sequences

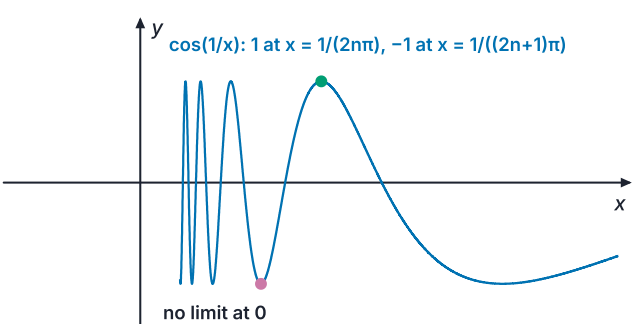
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How to show a quotient or a derivative has no limit.

MATHS TOOL

1. Find a sequence $x_n \to c$ ($x_n \neq c$) with $f(x_n) \to A$.
2. Find another $y_n \to c$ ($y_n \neq c$) with $f(y_n) \to B$ and $B \neq A$.
3. By the sequential criterion (P04) the limit does not exist, since a limit would force $A = B$.

Micro-example. $\cos \dfrac{1}{x}$ at 0: $x_n = \dfrac{1}{2n\pi}$ gives $\cos 2n\pi = 1$; $y_n = \dfrac{1}{(2n+1)\pi}$ gives -1. So no limit.



cos(1/x) takes the values 1 and −1 at points that crowd into 0, so it has no limit at 0. This is why a product law cannot be applied to a term like $x\cos(1/x)$ without a squeeze.

CHECK YOURSELF

Show $\sin\dfrac{1}{x}$ has no limit at 0 using $x_n = \dfrac{1}{n\pi}$ and one other sequence.

$x_n = \dfrac{1}{n\pi}$ gives $\sin n\pi = 0$. Take $y_n = \dfrac{1}{\pi/2 + 2n\pi}$: $\sin(\pi/2 + 2n\pi) = 1$. Limits $0 \neq 1$.

M08 Composite bookkeeping

index ↑

> Name the inside and the outside before you differentiate.

MATHS TOOL

1. Name the **inner** function f (the one applied first) and the **outer** function g.
2. Differentiate g, but evaluate g' at $f(x)$, **not at x.**
3. Multiply by $f'(x)$: $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$.

Micro-example. $\sqrt{x^2 - 2x + 5}$: inner $f = x^2 - 2x + 5$, outer $g(y) = \sqrt{y}$ with $g'(y) = \dfrac{1}{2\sqrt{y}}$. So the derivative is $\dfrac{1}{2\sqrt{x^2 - 2x + 5}} \cdot (2x - 2) = \dfrac{x - 1}{\sqrt{x^2 - 2x + 5}}$.

For three layers, peel from the outside: $\sin(\cos(x^2))$ gives $\cos(\cos x^2) \cdot (-\sin x^2) \cdot 2x$.

CHECK YOURSELF

Differentiate $\sin(x^2)$ and $(x^2+1)^3$.

$\cos(x^2)\cdot 2x$. And $3(x^2+1)^2\cdot 2x = 6x(x^2+1)^2$.

M09 Defining a function by cases, and the join

index ↑

The idea behind Carathéodory's theorem (T05).

MATHS TOOL

For a function f and a point c, define a new function by cases:

$\varphi(x) = \dfrac{f(x)-f(c)}{x-c}$ for $x \neq c$, and $\varphi(c) =$ a number we choose.

The only question is: **do the two descriptions meet at c?** That is, does $\lim_{x\to c}\varphi(x)$ exist and equal the number we chose? If yes, φ is continuous at c.

Micro-example. $f = x^2$, $c = 1$: $\varphi(x) = x+1$ for $x \neq 1$, and choosing $\varphi(1) = 2$ makes φ continuous. That chosen value is $f'(1)$.

Micro-example. $f = |x|$, $c = 0$: $\varphi(x) = \operatorname{sgn} x$ for $x \neq 0$, which jumps from -1 to $+1$. No value for $\varphi(0)$ can join them.

CHECK YOURSELF

For $f = x^3$ and any c, find φ (use M02) and say which value to give $\varphi(c)$.

$\varphi(x) = x^2 + xc + c^2$ for $x \neq c$. It is a polynomial, so choose $\varphi(c) = 3c^2$, which is $f'(c)$.

M10 "There exists c", and proof by contradiction

index ↑

How to read the conclusion of Rolle, MVT and Darboux.

MATHS TOOL

- "There exists at least one point c" says c exists. It does **not** say where, or that it is the only one, or that it is the midpoint.
- **Contradiction**: to show $f'(c) = 0$, assume $f'(c) > 0$ and derive something that is false; then do the same with $f'(c) < 0$.
- **Reduction by a minus sign**: a maximum of f is a minimum of $-f$. A proof for "max" gives the proof for "min" by replacing f with $-f$. Also $f'(c) = 0 \iff (-f)'(c) = 0$.

THE EXAM HABIT

When you finish a proof of "there exists c", say where c lies: "with $a < c < b$". And when you use "by symmetry" or "similarly", write one line showing how, because the examiner marks that line.

CHECK YOURSELF

Rolle gives c with $f'(c) = 0$ for $f = x(x-1)(x-2)$ on $[0,2]$. Is c unique?

No. $f' = 3x^2 - 6x + 2 = 0$ gives $x = 1 \pm \frac{1}{\sqrt{3}}$, both in $(0,2)$. "At least one" allows several.

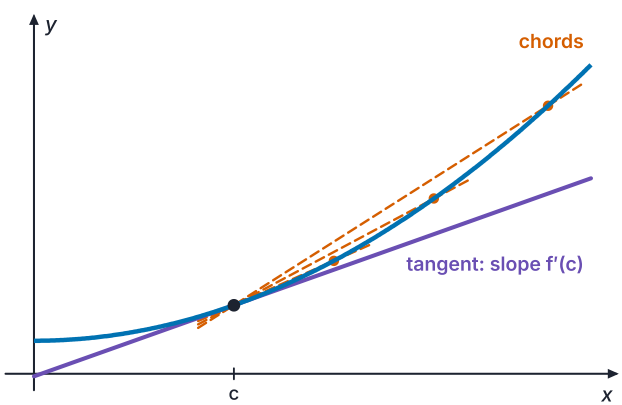
UNIT 2 · BARTLE §6.1

The derivative at a point

T01 The derivative: a limit of chord slopes

index ↑

> From the tangent of Plus Two to the ε–δ definition (related FYUGP CALC-24 Q5).



The derivative. The chord from (c, f(c)) to (x, f(x)) has slope (f(x) − f(c))/(x − c). As x → c the chords turn into the tangent; its slope is f′(c) = $\lim_{x \to c}$ (f(x) − f(c))/(x − c).

DEFINITION 6.1.1 · SAY IT WORD FOR WORD

Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that a real number L is the **derivative of f at c** if, given any $\varepsilon > 0$, there exists $\delta(\varepsilon) > 0$ such that if $x \in I$ satisfies $0 < |x - c| < \delta(\varepsilon)$, then

$$\left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon.$$

In this case we say f is **differentiable at c** and write $f'(c) = L$.

In plain words: the slope of the chord from $(c, f(c))$ to $(x, f(x))$ can be forced as close to L as anyone demands, just by taking x close enough to c (and not equal to c). Equivalent short form: $f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$. If c is an endpoint of I the limit is one-sided.

PLUS TWO SAID H, BARTLE SAYS X − C

Plus Two: $\frac{f(c+h) - f(c)}{h}$ as $h \to 0$. Put $h = x - c$ and it is the same limit. Why "0 < |x − c|"? At $x = c$ the quotient would be 0/0, so $x = c$ is never allowed.

THE GAME (SAME SHAPE AS CONTINUITY, M02 IN MODULE 1)

WHO	MOVES
Opponent	picks a band of half-height ε round L
You	pick a window δ round c (leaving out c itself)
Test	every chord slope inside the window lies in the band
δ may depend on	**ε, f and c**; it must **not** depend on x

SKELETON · PROVE F′(C) = L FROM THE DEFINITION

Let $\varepsilon > 0$. For $x \neq c$ simplify $\dfrac{f(x)-f(c)}{x-c} = $? . Choose $\delta =$? . If $0 < |x-c| < \delta$ then $\left|\dfrac{f(x)-f(c)}{x-c} - L\right| =$? $< \varepsilon$. Hence $f'(c) = L$. ∎

Know $f(x) = x^2$, $c = 2$, claim $L = 4$.

Want for every ε a δ with $\left|\dfrac{x^2-4}{x-2} - 4\right| < \varepsilon$ when $0 < |x-2| < \delta$.

SCRATCH · WORKED EXAMPLE 1 (NEVER COPIED)

For $x \neq 2$: $\dfrac{x^2-4}{x-2} = x+2$. So the distance to L is $|x+2-4| = |x-2|$.

Want $|x-2| < \varepsilon \Leftarrow |x-2| < \delta$ with $\delta = \varepsilon$.

PROOF · WHAT YOU WRITE

Let $\varepsilon > 0$ and put $\delta = \varepsilon$. If $0 < |x-2| < \delta$, then $\dfrac{x^2-4}{x-2} = x+2$, so $\left|\dfrac{x^2-4}{x-2} - 4\right| = |x-2| < \varepsilon$. Hence $f'(2) = 4$. ∎

Check: ε = 0.01, δ = 0.01, x = 2.005: quotient $= (4.020025 - 4)/0.005 = 4.005$, and $|4.005 - 4| = 0.005 < 0.01$.

In the model paper: **Q1** (this example).

WORKED EXAMPLE 2 · FILL THE GAPS (BARTLE EX 6.1.1(B), 1/X)

$g(x) = 1/x$, $c \neq 0$. For $x \neq c$, $x \neq 0$:
$\dfrac{1/x - 1/c}{x-c} = \dfrac{(c-x)/(xc)}{x-c} = ?$.

As $x \to c$ the quotient tends to ?. So $g'(c) = ?$.

Answer: the quotient is $\dfrac{-1}{xc}$, the limit is $-\dfrac{1}{c^2}$. (At c = 2: $-1/4$; at x = 2.5 the chord slope is $-1/5$, nearby.)

CHECK YOURSELF · CALC-24 Q5

From the definition, find $f'(c)$ for $f(x) = 3x$.

$\dfrac{3x-3c}{x-c} = 3$ for every $x \neq c$, so $|3-3| = 0 < \varepsilon$ with any δ. $f'(c) = 3$ for every c.

RESULT · SEQUENCE FORM (BARTLE EX 6.1.13)

If f is differentiable at c then $f'(c) = \lim_{n\to\infty} n\{f(c+\frac{1}{n}) - f(c)\}$. Reason: $x_n = c + \frac{1}{n} \to c$ with $x_n \neq c$, and $n\{f(x_n) - f(c)\} = \dfrac{f(x_n) - f(c)}{x_n - c}$ (sequential criterion for limits).

The converse is false. $f = |x|$, $c = 0$: $n\{\,|1/n| - 0\,\} = 1$, so the sequence converges to 1, yet $f'(0)$ does not exist (T02).

HYPOTHESES CHECKLIST (WHAT THE DEFINITION DEMANDS)

- ☐ the quotient has a limit
  $|x|/x$ is +1 on the right, −1 on the left: no limit **C5**
- ☐ that limit is a real number (finite)
  $x^{1/3}$ at 0: the quotient $x^{-2/3} \to \infty$, vertical tangent, no derivative
- ☐ c belongs to the interval I
  at an endpoint only a one-sided derivative makes sense

YOUR INTUITION WILL LIE TO YOU HERE

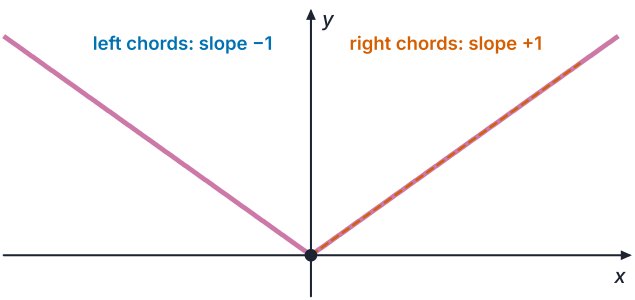
"The derivative is the slope of the tangent line, found by the rules." The rules (T03 to T08) hold only at points where the pieces are differentiable. At a joint or at a point where an inner piece is undefined (T09) you must go back to the limit of the quotient.

In the model paper: **Q2** (CALC-24 Q5), **Q3**, **Q4**.

T02 Differentiable ⇒ continuous (and not conversely)

index ↑

> The first theorem of the unit; the piecewise trap (related FYUGP CALC-24 Q13).



f(x) = |x| at 0: chords from the right have slope +1, from the left −1. The two one-sided limits differ, so f′(0) does not exist, although f is continuous **C5**. Continuous does not imply differentiable.

THEOREM 6.1.2

If $f : I \to \mathbb{R}$ has a derivative at $c \in I$, then f is continuous at c.

In plain words: no jumps and no holes at a point where there is a tangent. Contrapositive: if f is **not** continuous at c, then f′(c) does not exist. That is the quickest way to answer a piecewise question.

SKELETON

For $x \neq c$ write $f(x) - f(c) =$? $\cdot(x - c)$. Let $x \to c$: the first factor tends to ?, the second to ?. So $f(x) \to f(c)$. ∎

Know The quotient $\dfrac{f(x)-f(c)}{x-c}$ has limit $f'(c)$ as $x \to c$.

Want $\lim_{x\to c} f(x) = f(c)$, that is, $\lim(f(x)-f(c)) = 0$.

PROOF OF 6.1.2

1. For $x \neq c$: $f(x)-f(c) = \dfrac{f(x)-f(c)}{x-c}\,(x-c)$.
   Multiply and divide by $x-c \neq 0$. This is only algebra, and it needs $x \neq c$.

2. Let $x \to c$: the first factor tends to $f'(c)$, the second to 0.
   Hypothesis for the first factor; $x-c \to 0$ is trivial.

3. So $\lim_{x\to c}(f(x)-f(c)) = f'(c)\cdot 0 = 0$.
   Product of limits (Theorem 4.2.4): both limits exist.

4. Hence $\lim_{x\to c} f(x) = f(c)$, so f is continuous at c. ∎
   Definition of continuity at a cluster point (5.1.2).

HYPOTHESES CHECKLIST · "THE CONVERSE IS FALSE"

- ☐ f′(c) must exist
  $|x|$ is continuous at 0 but has no derivative there C5
- ☐ the derivative must be finite
  $x^{1/3}$ is continuous at 0, quotient $x^{-2/3}$; at $x_n = n^{-3}$ it equals $n^2 \to \infty$
- ☐ the same point c everywhere
  differentiable at c says nothing about continuity at other points (N4)

WHERE EACH HYPOTHESIS WAS USED

Step 2 uses the existence of the limit $f'(c)$ (a real number). Nothing else is needed: no continuity near c, no interval beyond c ∈ I.

WORKED EXAMPLE 1 · |X| AT 0

$\dfrac{|x|-|0|}{x-0} = \dfrac{|x|}{x} = \begin{cases} +1 & x>0 \\ -1 & x<0 \end{cases}$. Two sequences $x_n = \dfrac{1}{n}$ and $y_n = -\dfrac{1}{n}$ both tend to 0, but the quotient gives 1 and -1.

So no limit: $f'(0)$ does not exist, although $|x|$ is continuous. C5 (The same quotient is the old MCQ "$\lim_{x\to 0} |x|/x$": answer, no limit.)

WORKED EXAMPLE 2 · PIECEWISE AT A JOINT (CALC-24 Q13, FILL THE GAPS)

$f(x) = x^2$ for $x \le 1$, $f(x) = 4\sqrt{x}$ for $x > 1$. Is f differentiable at 1?

$f(1) =$?. $\lim_{x \to 1^+} 4\sqrt{x} =$?. Equal? ? So f is ? at 1, hence by 6.1.2 ?.

Answer: $f(1) = 1$, right limit $= 4$. Not equal, so f is discontinuous at 1 and $f'(1)$ does **not** exist. The question "if so, find the value" is a trap.

Making it work: $F(x) = x^2$ $(x \le 1)$, $ax + b$ $(x > 1)$. Continuity: $a + b = 1$. Left derivative: $\dfrac{x^2 - 1}{x - 1} = x + 1 \to 2$. Right derivative (with $b = 1 - a$): $\dfrac{a(x-1)}{x-1} = a$. So $a = 2$, $b = -1$, $F'(1) = 2$.

CHECK YOURSELF

$G(x) = x^2$ for $x \le 1$, $G(x) = \sqrt{x}$ for $x > 1$. Differentiable at 1?

Continuous: both sides give 1. Left derivative: $x + 1 \to 2$. Right derivative: $\dfrac{\sqrt{x} - 1}{x - 1} = \dfrac{1}{\sqrt{x} + 1} \to \dfrac{1}{2}$. Different, so **not** differentiable at 1.

EXAM TRAP · PIECEWISE FUNCTIONS

Always in this order: (1) values equal at the joint? (continuity); (2) one-sided derivatives equal? Matching values alone are not enough (G above); matching values **and** slopes give differentiability.

In the model paper: **Q5** ($x^{1/3}$), **Q21**, **Q22** (CALC-24 Q13).

T03 Algebra of derivatives: sum, product, quotient

index ↑

> The rules you know from Plus Two, now with proofs (Bartle 6.1.3).

PICTURE · A RECTANGLE THAT GROWS

Let f be the width and g the height. Nudge x: the width grows by Δf and the height by Δg. The new area is the old one plus a strip $g\,\Delta f$, plus a strip $f\,\Delta g$, plus a tiny corner $\Delta f\,\Delta g$. Divide by Δx : the corner disappears because $\Delta f \to 0$. That is $(fg)' = f'g + fg'$.

THEOREM 6.1.3

Let $f, g : I \to \mathbb{R}$ be differentiable at $c \in I$. Then:
(a) for $\alpha \in \mathbb{R}$, $(\alpha f)'(c) = \alpha f'(c)$;
(b) $(f+g)'(c) = f'(c) + g'(c)$;
(c) **Product Rule**: $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$;
(d) **Quotient Rule**: if $g(c) \neq 0$, then $\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}$.

In plain words: differentiate one piece at a time. Product: "first-prime second plus first second-prime". Quotient: "bottom times top-prime, minus top times bottom-prime, over bottom squared".

SKELETON · PRODUCT RULE

For $x \neq c$: $\dfrac{f(x)g(x) - f(c)g(c)}{x - c} = \dfrac{f(x) - f(c)}{x - c} \cdot$? $+$? $\cdot \dfrac{g(x) - g(c)}{x - c}$. Let $x \to c$ using that g is ? at c. ∎

PROOF OF (A), (B), (C), (D)

(a), (b). $\dfrac{(\alpha f)(x)-(\alpha f)(c)}{x-c} = \alpha\dfrac{f(x)-f(c)}{x-c}$, and the quotient of a sum is the sum of quotients. Apply the limit laws (4.2.4).

(c)

1. For $x \neq c$: $\dfrac{f(x)g(x)-f(c)g(c)}{x-c} = \dfrac{f(x)-f(c)}{x-c}\,g(x) + f(c)\,\dfrac{g(x)-g(c)}{x-c}$.
   Add and subtract $f(c)g(x)$ in the numerator. This one trick is the whole proof.

2. g is continuous at c, so $g(x) \to g(c)$.
   Theorem 6.1.2 (T02): g is differentiable at c. Without it the first term has no limit.

3. Let $x \to c$: the limit is $f'(c)g(c) + f(c)g'(c)$. ∎
   Limit laws: the four factors have limits $f'(c)$, $g(c)$, $f(c)$, $g'(c)$.

(d)

1. Since g is continuous at c and $g(c) \neq 0$, $g \neq 0$ on a neighbourhood J of c.
   Local sign (Theorem 4.2.9). This is what makes f/g defined near c.

2. For $x \in J$, $x \neq c$: $\dfrac{f(x)/g(x)-f(c)/g(c)}{x-c} = \dfrac{1}{g(x)g(c)}\left[\dfrac{f(x)-f(c)}{x-c}\,g(c) - f(c)\,\dfrac{g(x)-g(c)}{x-c}\right]$.
   Common denominator, then add and subtract $f(c)g(c)$.

3. Let $x \to c$ (using $g(x) \to g(c)$): the limit is $\dfrac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$. ∎

Limit laws.

HYPOTHESES CHECKLIST

- ☐ both f and g differentiable at c
  $|x| \cdot |x| = x^2$ is differentiable at 0 although $|x|$ is not: the rule does not apply, and the converse is false **C5**
- ☐ $g(c) \neq 0$ for the quotient
  $f = 1$, $g = x$ at 0: $1/x$ is not even defined at 0
- ☐ same point c, and f, g both on I
  pieces defined on different intervals cannot be combined

EXAM TRAP

$(fg)' \neq f'g'$. With $f = g = x$: $(x^2)' = 2x$, but $f'g' = 1$. Likewise $(f/g)' \neq f'/g'$.

WORKED EXAMPLE 1 · QUOTIENT RULE (BARTLE EX 6.1.5(A); RELATED CALC-24)

$f(x) = \dfrac{x}{1+x^2}$. Top $u = x$, $u' = 1$; bottom $v = 1 + x^2 \geq 1 \neq 0$, $v' = 2x$.

$$f'(x) = \frac{1 \cdot (1+x^2) - x \cdot 2x}{(1+x^2)^2} = \frac{1 - x^2}{(1+x^2)^2}.$$

Check: $f'(0) = 1$ and $f'(1) = 0$ (the top of the hump, $f(1) = \dfrac{1}{2}$).

WORKED EXAMPLE 2 · PRODUCT RULE, FILL THE GAPS

$h(x) = (2x+1)(x^2-3)$. Let $u = 2x+1$, $u' = ?$; $v = x^2-3$, $v' = ?$. Then $h' = u'v + uv' = ?$.

Answer: $u' = 2$, $v' = 2x$, $h' = 2(x^2-3) + (2x+1)2x = 6x^2 + 2x - 6$. Check by expanding: $2x^3 + x^2 - 6x - 3$ differentiates to the same.

CHECK YOURSELF

Find $\left(\dfrac{x^2+1}{x-2}\right)'$ at $x = 3$. Where is it not defined?

$\dfrac{2x(x-2) - (x^2+1)}{(x-2)^2} = \dfrac{x^2 - 4x - 1}{(x-2)^2}$. At 3: $\dfrac{9 - 12 - 1}{1} = -4$. Not defined at $x = 2$ ($g(2) = 0$).

In the model paper: **Q6**, **Q23** (also the special case $(1/g)' = -g'/g^2$, Bartle Ex 6.1.7(b), which needs only $g(c) \neq 0$).

T04 Sums, products of many functions; the power rule

index ↑

Induction on the product rule (Bartle 6.1.4).

PICTURE · A BOX WITH N SIDES

A product of n factors changes by letting **one factor at a time** change while the others stand still: n terms, each with one prime.

COROLLARY 6.1.4

Let $f_1, \dots, f_n$ be functions on I, differentiable at $c \in I$.
(a) $(f_1 + \cdots + f_n)'(c) = f_1'(c) + \cdots + f_n'(c)$.
(b) $(f_1 f_2 \cdots f_n)'(c) = f_1'(c) f_2(c) \cdots f_n(c) + \cdots + f_1(c) \cdots f_{n-1}(c) f_n'(c)$.
In particular, $(f^n)'(c) = n\,(f(c))^{n-1} f'(c)$, and taking $f(x) = x$: $D\,x^n = n x^{n-1}$ for $n \in \mathbb{N}$.

In plain words: bring the power down, lower it by one, and multiply by the derivative of what is inside (the last factor is the one students forget).

SKELETON · INDUCTION

Base $n = 1$: ? . Step: assume true for n; write $f_1 \cdots f_{n+1} = (f_1 \cdots f_n) \cdot f_{n+1}$ and apply the ? rule. ∎

PROOF (POWER RULE BY INDUCTION)

1. $n = 1$: $D\,x = 1 = 1 \cdot x^0$.
   The quotient $\dfrac{x - c}{x - c} = 1$.

2. Assume $D\,x^k = k x^{k-1}$. Write $x^{k+1} = x^k \cdot x$.
   So the product rule (T03(c)) applies with $f = x^k$, $g = x$, both differentiable.

3. $D\,x^{k+1} = k x^{k-1} \cdot x + x^k \cdot 1 = (k+1) x^k$. ∎
   Add the two terms: $k x^k + x^k$.

RESULT · NEGATIVE POWERS (QUOTIENT RULE)

$x^{-n} = 1/x^n$ with $g(x) = x^n \neq 0$ for $x \neq 0$: $D\,x^{-n} = \dfrac{0 - nx^{n-1}}{x^{2n}} = -nx^{-n-1}$. At $x = 2$, $n = 3$: $-3/16 = -0.1875$.

HYPOTHESES CHECKLIST

- ☐ n is a natural number (finite product)
  real powers x^r come later (Bartle §8.3); quote $D\,x^r = rx^{r-1}$ as a known fact
- ☐ each f_i differentiable at c
  $f = |x|$: $f^2 = x^2$ has derivative 0 at 0, but $2ff'$ has no meaning because $f'(0)$ does not exist C5

EXAM TRAP

$(f^n)' = nf^{n-1}f'$, not nf^{n-1}. Test with $f = x^2$, $n = 3$: x^6 has derivative $6x^5$; "$3f^2$" gives $3x^4$ only, and the missing factor $f' = 2x$ gives $3x^4 \cdot 2x = 6x^5$.

CHECK YOURSELF

Differentiate $(x^2 + 1)^3$ and evaluate at $x = 1$.

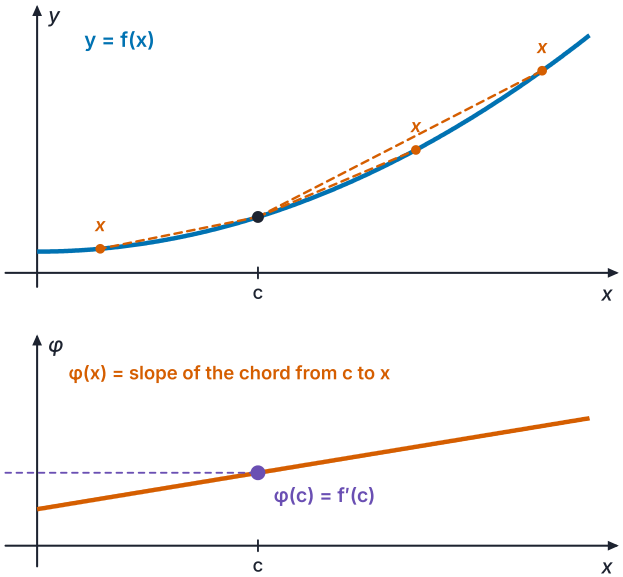
$3(x^2 + 1)^2 \cdot 2x = 6x(x^2 + 1)^2$. At 1: $6 \cdot 4 = 24$.

In the model paper: **Q7**.

T05 Carathéodory's characterisation

index ↑

> "Fill the hole in the chord-slope" (Bartle 6.1.5); the tool that proves the chain rule.



Carathéodory's idea ($f(x) = x^2/4 + 0.3$, $c = 1.4$). The chord-slope function $\varphi(x) = (f(x) - f(c))/(x - c)$ has a hole at $x = c$. f is differentiable at c exactly when that hole can be filled so that φ is **continuous** at c, and then $\varphi(c) = f'(c)$. Here $\varphi(x) = (x + c)/4$.

THEOREM 6.1.5 · CARATHÉODORY

Let f be defined on an interval I containing the point c. Then f is differentiable at c if and only if there exists a function φ on I that is continuous at c and satisfies

$$f(x) - f(c) = \varphi(x)\,(x - c) \quad \text{for } x \in I.$$

In this case, $\varphi(c) = f'(c)$.

In plain words: write the difference $f(x) - f(c)$ as "something times $(x - c)$". Differentiable means you can choose that "something" so that it behaves nicely (is continuous) at c. Its value at c is the derivative.

SKELETON

(⇒) Define $\varphi(x) =$? for $x \neq c$ and $\varphi(c) =$?. Check continuity at c and the identity at every x. (⇐) For $x \neq c$ divide to get $\dfrac{f(x)-f(c)}{x-c} =$?; let $x \to c$. ∎

PROOF OF 6.1.5

(⇒) Assume $f'(c)$ exists.

1. Define $\varphi(x) = \dfrac{f(x) - f(c)}{x - c}$ for $x \neq c$ and $\varphi(c) = f'(c)$.
   The difference quotient is the only possible choice for $x \neq c$; the hole at c is filled with the derivative.

2. φ is continuous at c: $\lim_{x \to c} \varphi(x) = f'(c) = \varphi(c)$.
   Definition of $f'(c)$.

3. $f(x) - f(c) = \varphi(x)(x - c)$ for every $x \in I$.
   For $x \neq c$ multiply out. For $x = c$ both sides are 0, so check c too.

(⇐) Assume φ exists, continuous at c.

1. For $x \neq c$: $\dfrac{f(x) - f(c)}{x - c} = \varphi(x)$.
   Divide by $x - c \neq 0$.

2. $\lim_{x \to c} \varphi(x) = \varphi(c)$.
   φ is continuous at c.

3. So the quotient has a limit: $f'(c)$ exists and equals $\varphi(c)$. ∎
   Definition 6.1.1.

HYPOTHESES CHECKLIST

- ☐ φ continuous **at c** (not everywhere)
  $|x|$ at 0: $\varphi(x) = \operatorname{sgn} x$ for $x \neq 0$ is forced, with no continuous value at 0 **C5**
- ☐ the identity holds for **all** $x \in I$, including c
  φ is determined for $x \neq c$ by the quotient; only its value at c is free
- ☐ I an interval containing c
  φ must be defined on all of I, otherwise the chain rule argument cannot compose

WORKED EXAMPLE 1 · $f(x) = x^3$ (BARTLE'S ILLUSTRATION)

$x^3 - c^3 = (x^2 + cx + c^2)(x - c)$. So $\varphi(x) = x^2 + cx + c^2$, a polynomial, continuous everywhere.

$f'(c) = \varphi(c) = c^2 + c^2 + c^2 = 3c^2$. Check at $c = 1$: $\varphi(x) = x^2 + x + 1$, $\varphi(1) = 3$, and $\varphi(2) = 7 = \dfrac{8-1}{2-1}$.

WORKED EXAMPLE 2 · $\sqrt{x}$ AT $c > 0$, FILL THE GAPS

$\sqrt{x} - \sqrt{c} = \dfrac{x - c}{\sqrt{x} + \sqrt{c}}$ (multiply top and bottom by $\sqrt{x} + \sqrt{c}$). So $\varphi(x) =$? for $x \neq c$, and $\varphi(c) =$?.

Answer: $\varphi(x) = \dfrac{1}{\sqrt{x} + \sqrt{c}}$, continuous at c since $\sqrt{x} + \sqrt{c} \to 2\sqrt{c} \neq 0$ (this needs $c > 0$). $f'(c) = \dfrac{1}{2\sqrt{c}}$; at $c = 4$: $\dfrac{1}{4}$. At $c = 0$ there is no φ ($\varphi(x) = x^{-1/2}$ blows up).

CHECK YOURSELF

Find φ for $f(x) = x^2 + 3x$ at c and deduce $f'(c)$.

$f(x) - f(c) = (x^2 - c^2) + 3(x - c) = (x + c + 3)(x - c)$. So $\varphi(x) = x + c + 3$ and $f'(c) = 2c + 3$.

YOUR INTUITION WILL LIE TO YOU HERE

"φ is the derivative function." No: φ depends on the point c **and** on x. It is the chord slope from c, whose value **at x = c** is $f'(c)$.

In the model paper: **Q24**.

T06 The chain rule

index ↑

> Derivative of a composite (Bartle 6.1.6; 10-mark proof).

PICTURE · GEARS

If y changes 3 times as fast as u, and u changes 2 times as fast as x, then y changes 6 times as fast as x. The rates multiply. The only care: the outer rate is measured at the **inner value** $f(c)$, not at c.

THEOREM 6.1.6 · CHAIN RULE

Let I, J be intervals in $\mathbb{R}$, let $g : I \to \mathbb{R}$ and $f : J \to \mathbb{R}$ be functions such that $f(J) \subseteq I$, and let $c \in J$. If f is differentiable at c and g is differentiable at $f(c)$, then the composite $g \circ f$ is differentiable at c and

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c).$$

In plain words: differentiate the outside (leave the inside alone), then multiply by the derivative of the inside.

WHY THE QUICK PROOF FAILS (N1)

Quick idea: $\frac{g(f(x)) - g(f(c))}{x - c} = \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \cdot \frac{f(x) - f(c)}{x - c}$. But $f(x) = f(c)$ may happen at points x arbitrarily close to c (for $x^2 \sin(1/x)$ at 0, $f(\frac{1}{n\pi}) = 0$), so you would divide by zero. Carathéodory avoids all division.

SKELETON

Get φ for f at c. Get ψ for g at $d = f(c)$. Substitute $y = f(x)$: $g(f(x)) - g(f(c)) =$? $\cdot(x - c)$. Show the bracket is continuous at c with value ? . ∎

PROOF OF 6.1.6

1. There is φ on J, continuous at c, with $f(x) - f(c) = \varphi(x)(x - c)$ and $\varphi(c) = f'(c)$.
   Carathéodory (⇒) for f at c (T05).

2. With $d = f(c)$ there is ψ on I, continuous at d, with $g(y) - g(d) = \psi(y)(y - d)$ and $\psi(d) = g'(d)$.
   Carathéodory again, for the outer function g at d.

3. Put $y = f(x)$ ($x \in J$, allowed since $f(J) \subseteq I$): $g(f(x)) - g(f(c)) = \psi(f(x))\,[f(x) - f(c)] = \big[\psi(f(x))\,\varphi(x)\big](x - c)$.
   Substitute the identity of step 2, then of step 1. Pure algebra, no division.

4. $\Phi(x) = \psi(f(x))\,\varphi(x)$ is continuous at c.
   f is continuous at c (6.1.2); ψ is continuous at $f(c)$, so $\psi \circ f$ is continuous at c (composite theorem, Module 1); φ is continuous at c; products of continuous functions are continuous.

5. $\Phi(c) = \psi(f(c))\varphi(c) = g'(f(c))\,f'(c)$.
   $\psi(d) = g'(d)$ and $\varphi(c) = f'(c)$.

6. By Carathéodory (⇐), $g \circ f$ is differentiable at c with $(g \circ f)'(c) = \Phi(c)$. ∎
   Step 3 is exactly the Carathéodory identity for $g \circ f$.

Marks (10): statement 2 · Carathéodory for f 2 · for g 2 · composition identity 2 · continuity of Φ and conclusion 2.

HYPOTHESES CHECKLIST

☐ f differentiable at c

$f = |x|$ at 0 with $g(y) = y^2$: $(g \circ f)(x) = x^2$ happens to work, but $2|x|\ f'$ is meaningless

☐ g differentiable at the point $f(c)$, not at c

$g = |y|$, $f(x) = x$ at $c = 0$: $g \circ f = |x|$ has no derivative C5

☐ $f(J) \subseteq I$ (composite defined)

$g = \sqrt{y}$ on $[0, \infty)$, $f(x) = x - 1$ on $\mathbb{R}$: $g(f(x))$ is undefined for $x < 1$

NOTE

The hypotheses are sufficient, not necessary: $g = |y|$, $f = x^2$ at 0 gives $g \circ f = x^2$, differentiable although g is not differentiable at $f(0) = 0$.

WORKED EXAMPLE 1 · BARTLE EX 6.1.5(B)

$h(x) = \sqrt{x^2 - 2x + 5}$. Name the inner $f(x) = x^2 - 2x + 5$ and outer $g(y) = \sqrt{y}$.

Inner value $f(x) = (x-1)^2 + 4 \geq 4 > 0$, so g is differentiable at $f(x)$ for every x.

$g'(y) = \dfrac{1}{2\sqrt{y}}$ evaluated at $y = f(x)$, times $f'(x) = 2x - 2$:

$$h'(x) = \frac{2x - 2}{2\sqrt{x^2 - 2x + 5}} = \frac{x - 1}{\sqrt{x^2 - 2x + 5}}.$$

Check at $x = 3$: $\dfrac{2}{\sqrt{8}} = 0.7071$.

WORKED EXAMPLE 2 · FILL THE GAPS

$k(x) = (x^2+1)^{10}$. Inner $f = ?$, outer $g(y) = ?$. $k'(x) = g'(f(x))f'(x) = ?$.

Answer: $f = x^2+1$, $g = y^{10}$; $k'(x) = 10(x^2+1)^9 \cdot 2x = 20x(x^2+1)^9$. At $x = 1$: $20 \cdot 2^9 = 10240$.

RESULT · EVEN AND ODD (BARTLE EX 6.1.9)

If f is differentiable and **even** ($f(-x) = f(x)$) then f' is **odd**; if f is **odd** then f' is **even**. Proof: differentiate $f(-x) = f(x)$ with the chain rule (inner $-x$, derivative -1): $-f'(-x) = f'(x)$, i.e. $f'(-x) = -f'(x)$. For odd f: $-f'(-x) = -f'(x)$, i.e. $f'(-x) = f'(x)$. Check: $x^2 \to 2x$ (odd); $x^3 \to 3x^2$ (even).

CHECK YOURSELF

Differentiate $\tan(x^2)$ for $|x| < \sqrt{\pi/2}$, and evaluate at $x = 0$.

Outer $\tan y$, $D\tan y = \sec^2 y$ (T08), inner x^2: $2x\sec^2(x^2)$. At 0: 0.

In the model paper: Q8 , Q9 , Q37 .

T07 The derivative of |f| and of sgn

index ↑

> Where absolute-value functions have corners (Bartle Example 6.1.7(c), Ex 6.1.7, 6.1.8).

PICTURE · FOLDING THE GRAPH

$|f|$ is the graph of f with every part below the x-axis folded up. Smooth parts stay smooth; only where f crosses the axis with a non-zero slope does the fold make a **corner**. Example $|x^2 - 1|$: corners (cusps) at $(\pm 1, 0)$, a hump of height 1 at $x = 0$, and the two sides rising like $x^2 - 1$ for $|x| > 1$.

EXAMPLE 6.1.7(C) AND EXERCISE 6.1.7

$g(x) = |x|$ is differentiable at every $x \neq 0$ with $g'(x) = \operatorname{sgn} x$ (where $\operatorname{sgn} x = 1, 0, -1$ for $x > 0, = 0, < 0$); $g'(0)$ does not exist. Consequently, if f is differentiable at c and $f(c) \neq 0$, then

$$|f|'(c) = \operatorname{sgn}(f(c))\, f'(c).$$

If $f(c) = 0$, then $|f|$ is differentiable at c **if and only if** $f'(c) = 0$.

In plain words: away from zeros, $|f|$ has slope $+f'$ where f is positive and $-f'$ where f is negative. At a zero of f the fold is smooth only if f is already flat there.

SKELETON · $f(c) = 0$

For $x \neq c$: $\dfrac{|f(x)| - 0}{x - c} = \pm \left| \dfrac{f(x)}{x - c} \right|$ with sign ? for $x > c$ and ? for $x < c$. Right limit = ? , left limit = ? . Equal iff ? . ∎

PROOF (F(C) = 0 CASE)

1. For $x > c$: $\dfrac{|f(x)| - |f(c)|}{x - c} = \dfrac{|f(x)|}{x - c} = \left|\dfrac{f(x)}{x - c}\right|$. For $x < c$: it equals $-\left|\dfrac{f(x)}{x - c}\right|$.

   $f(c) = 0$; the sign of $x - c$ is + on the right and − on the left, and $|x - c|$ is the absolute value of the denominator.

2. As $x \to c$, $\left|\dfrac{f(x)}{x - c}\right| \to |f'(c)|$.

   $\dfrac{f(x)}{x - c} = \dfrac{f(x) - f(c)}{x - c} \to f'(c)$, and $|\cdot|$ is continuous.

3. So the right limit is $+|f'(c)|$ and the left limit is $-|f'(c)|$.

   Step 1 and 2 combined.

4. They agree iff $|f'(c)| = 0$, that is $f'(c) = 0$. ∎

   One-sided limits must be equal for the limit to exist (M03, Module 1).

HYPOTHESES CHECKLIST

- ☐ f differentiable at c

  $f = |x|$ itself: $||x|| = |x|$, no derivative at 0 **C5**

- ☐ $f(c) \neq 0$ for the formula $\operatorname{sgn}(f)f'$

  $|\sin x|$ at 0: the formula gives $\operatorname{sgn}(0)\cos 0 = 0$, but the true answer is "no derivative"

- ☐ zero of f needs $f'(c) = 0$

  $f = x^2$ at 0: $f' = 0$, and $|x^2| = x^2$ is differentiable.

  $f = \sin x$ at $k\pi$: $f' = \pm 1 \neq 0$, corner

YOUR INTUITION WILL LIE TO YOU HERE

"$|f|$ has a corner wherever $f = 0$." False: $|x^2| = x^2$ has none. A corner appears only if the zero is a **crossing** ($f' \neq 0$ there).

WORKED EXAMPLE 1 · $|x^2 - 1|$ (BARTLE EX 6.1.7(C) FIGURE)

$f = x^2 - 1$, $f' = 2x$, zeros at ± 1 with $f'(\pm 1) = \pm 2 \neq 0$.

Away from ± 1: $|f|' = \operatorname{sgn}(x^2 - 1) \cdot 2x$, i.e. $2x$ for $|x| > 1$ and $-2x$ for $|x| < 1$. At ± 1: **not differentiable**.

Check: at $x = 2$, $|f| = 3$ and slope 4; at $x = 0.5$, $|f| = 0.75$ and slope -1 (the graph falls from the hump at 0 toward the cusp at 1).

WORKED EXAMPLE 2 · $h(x) = x|x|$ (EX 6.1.8(C)), FILL THE GAPS

For $x > 0$: $h =$?, $h' =$?. For $x < 0$: $h =$?, $h' =$?. At 0: $\dfrac{h(x) - h(0)}{x} =$? $\to$?.

Answer: $x^2 \to 2x$; $-x^2 \to -2x$; $\dfrac{x|x|}{x} = |x| \to 0$. So h is differentiable everywhere with $h'(x) = 2|x|$.

CHECK YOURSELF

Is $|\sin x|$ differentiable at $x = \pi$? At $x = \pi/2$?

At π: $\sin \pi = 0$ and $\cos \pi = -1 \neq 0$: **not** differentiable (corner). At $\pi/2$: $\sin = 1 \neq 0$, derivative $\operatorname{sgn}(1)\cos(\pi/2) = 0$.

In the model paper: Q10 , Q11 .

T08 Derivatives of tan, sec, cot, csc

index ↑

The quotient rule applied to sin and cos (Bartle Example 6.1.7(d)).

PICTURE

$\tan x$ climbs between its vertical asymptotes at $\pm\pi/2$; its slope is never below 1 (it is $\sec^2 x \geq 1$) and shoots to infinity near the asymptotes. At $x = \pi/4$ the slope is $\sec^2 \dfrac{\pi}{4} = 2$.

EXAMPLE 6.1.7(D) · USE $\sin' = \cos$, $\cos' = -\sin$ (PLUS TWO FACTS)

$D \tan x = \sec^2 x$ and $D \sec x = \sec x \tan x$ for $x \neq (2k+1)\pi/2$;
$D \cot x = -\csc^2 x$ and $D \csc x = -\csc x \cot x$ for $x \neq k\pi$.

Rule of thumb: the "co-" functions get a minus sign.

SKELETON

Write the function as a quotient: $\tan x = \dfrac{\sin x}{\cos x}$.
Apply (d) of Theorem 6.1.3, valid where the bottom is ? . Simplify using ? . ∎

PROOF

1. $\tan x = \sin x / \cos x$, with $\cos x \neq 0$ for $x \neq (2k+1)\pi/2$.
   Quotient rule needs $g(c) \neq 0$.

2. $D \tan x = \dfrac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \dfrac{\cos^2 x + \sin^2 x}{\cos^2 x} = \sec^2 x.$
   Theorem 6.1.3(d); then $\sin^2 + \cos^2 = 1$.

3. $D \sec x = D \dfrac{1}{\cos x} = \dfrac{0 \cdot \cos x - 1 \cdot (-\sin x)}{\cos^2 x} = \dfrac{\sin x}{\cos^2 x} = \sec x \tan x$. ∎
   $f = 1$, $f' = 0$ in the quotient rule.

HYPOTHESES CHECKLIST

☐ $\cos x \neq 0$ (for tan, sec) or $\sin x \neq 0$ (for cot, csc)
at $x = \pi/2$ the function $\tan x$ is not defined, so there is nothing to differentiate

☐ $\sin' = \cos$, $\cos' = -\sin$ known
Bartle proves these later; in the exam quote them as standard Plus Two facts

WORKED EXAMPLE 1 · COT

$\cot x = \cos x / \sin x$: $D\cot x = \dfrac{-\sin x \cdot \sin x - \cos x \cdot \cos x}{\sin^2 x} = \dfrac{-1}{\sin^2 x} = -\csc^2 x$, for $x \neq k\pi$.

Check at $x = \pi/4$: $-\csc^2 \dfrac{\pi}{4} = -(\sqrt{2})^2 = -2$; and numerically $\cot 0.8 = 0.9712$, $\cot 0.7854 = 1$, slope $\approx (0.9712 - 1)/0.0146 = -1.97$.

WORKED EXAMPLE 2 · CSC, FILL THE GAPS

$\csc x = 1/\sin x$. $D\csc x = \dfrac{0 \cdot \sin x - 1 \cdot (\ ?\ )}{\sin^2 x} =$?.

Answer: $\dfrac{-\cos x}{\sin^2 x} = -\csc x \cot x$.

CHECK YOURSELF

Find $\dfrac{d}{dx}(x \tan x)$ and evaluate at $x = \pi/4$.

Product rule: $\tan x + x \sec^2 x$. At $\pi/4$: $1 + \dfrac{\pi}{4} \cdot 2 = 1 + \dfrac{\pi}{2} \approx 2.571$.

EXAM TRAP

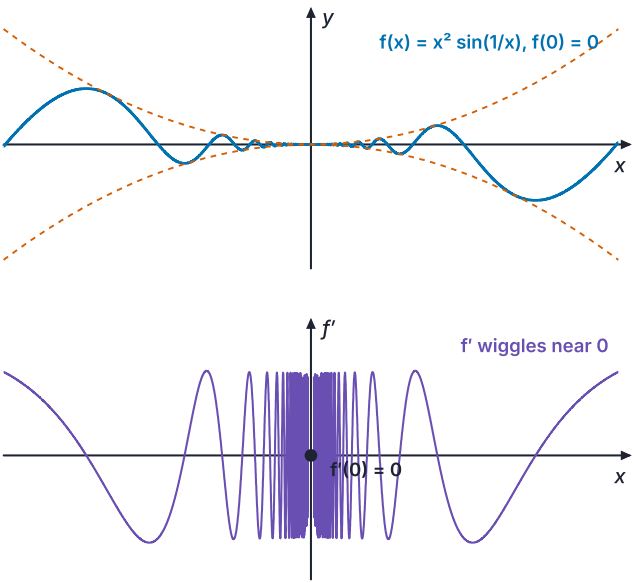
$D\sec x = \sec x \tan x$ (positive); $D\csc x = -\csc x \cot x$ (negative). And $D\tan x = \sec^2 x$, **not** $\sec x$ and not $1/\cos^2$ left unsimplified without saying so.

In the model paper: **Q12** .

T09 $x^2 \sin(1/x)$: differentiable, but f′ is not continuous

index ↑

> The most famous example of the unit (Bartle Example 6.1.7(e), Ex 6.1.10, 6.1.12).



$x^2 \sin(1/x)$ is differentiable everywhere ($f'(0) = 0$, squeezed by $\pm x^2$), but f' is **not continuous** at 0 C6. Even so, by Darboux, f' still takes every value between any two of its values: a derivative can wiggle, but it can never **jump**.

EXAMPLE 6.1.7(E)

Let $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Then f is differentiable at every $x \in \mathbb{R}$, with

$$f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x} \ (x \neq 0), \qquad f'(0) = 0,$$

but f' has no limit at 0, so f' is **not continuous** at 0.

In plain words: a function can have a derivative everywhere and still have a derivative that wiggles with the same height ± 1 no matter how close you get to 0. The tangent exists at 0 (it is the x-axis), but the slopes next to 0 do not settle on 0.

SKELETON

(1) For $x \neq 0$: product and chain rule. (2) At 0 use ? (rules do not apply). Quotient $\dfrac{f(x) - 0}{x} =$?; squeeze with $|\cdot| \leq$?. (3) Two sequences $x_n \to 0$ with $f'(x_n) \to$? and ?. ∎

PROOF

1. For $x \neq 0$: $f'(x) = 2x \sin \dfrac{1}{x} + x^2 \cos \dfrac{1}{x} \cdot \left(-\dfrac{1}{x^2}\right) = 2x \sin \dfrac{1}{x} - \cos \dfrac{1}{x}$.
   Product rule, with the chain rule for $\sin(1/x)$ (inner $1/x$, derivative $-1/x^2$). Valid because $1/x$ is differentiable at $x \neq 0$.

2. At 0: $\dfrac{f(x) - f(0)}{x - 0} = x \sin \dfrac{1}{x}$.
   Definition 6.1.1. The rules cannot be used: $\sin(1/x)$ is not even defined at 0 (N2).

3. $\left| x \sin \dfrac{1}{x} \right| \leq |x| \to 0$, so $f'(0) = 0$.
   $|\sin| \leq 1$ and squeeze: for ε take $\delta = \varepsilon$.

4. Let $x_n = \dfrac{1}{2n\pi}$: $f'(x_n) = 2x_n \cdot 0 - \cos 2n\pi = -1$. Let $y_n = \dfrac{1}{(2n+1)\pi}$: $f'(y_n) = 0 - \cos(2n+1)\pi = +1$.
   $\sin(2n\pi) = \sin(2n+1)\pi = 0$; $\cos 2n\pi = 1$; $\cos(2n+1)\pi = -1$.

5. $x_n \to 0$, $y_n \to 0$, but $f'(x_n) \to -1$ and $f'(y_n) \to +1$. So $\lim_{x \to 0} f'(x)$ does not exist; in particular it is not $f'(0) = 0$. ∎
   Two sequences with different limits (M07); discontinuity criterion 5.1.4.

Check: n = 1: $x_1 = \dfrac{1}{2\pi} = 0.159$, $f'(x_1) = -1$; $y_1 = \dfrac{1}{3\pi} = 0.106$, $f'(y_1) = +1$. Also $|f'(x)| \leq 2|x| + 1 \leq 3$ on $[-1, 1]$: bounded.

WHAT CAN AND CANNOT BE USED HERE

☐ rules (product, chain) at $x \neq 0$
fine: $1/x$ is differentiable there

☐ rules at $x = 0$
$\sin(1/x)$ is undefined at 0: you must use the limit of the quotient (N2)

☐ "differentiable ⇒ f' continuous"
false: this example **C6**

RESULT · WHICH POWERS WORK (BARTLE EX 6.1.12)

$f(x) = x^r \sin(1/x)$, $f(0) = 0$, $r > 0$. The quotient at 0 is $x^{r-1}\sin(1/x)$.

$r > 1$	$\lvert x^{r-1}\sin\frac{1}{x}\rvert \leq \lvert x\rvert^{r-1} \to 0$: $f'(0) = 0$ exists.
$r = 1$	quotient $\sin\frac{1}{x}$, no limit (two sequences): $f'(0)$ does not exist.
$0 < r < 1$	$\lvert x\rvert^{r-1} \to \infty$, unbounded: no derivative.

WORKED EXAMPLE 1 · DONE ABOVE; EXTENSION $x^2\sin(1/x^2)$ (EX 6.1.10, C19)

$g(x) = x^2\sin(1/x^2)$, $g(0) = 0$. At 0: quotient $x\sin(1/x^2) \to 0$. For $x \neq 0$: $g'(x) = 2x\sin\frac{1}{x^2} - \frac{2}{x}\cos\frac{1}{x^2}$.

At $x_n = \frac{1}{\sqrt{2n\pi}}$: $\sin\frac{1}{x_n^2} = \sin 2n\pi = 0$, $\cos = 1$, so $g'(x_n) = -2\sqrt{2n\pi}$. For $n = 1$: $-2\sqrt{2\pi} = -5.013$; for $n = 8$: $-2\sqrt{16\pi} = -14.18$. Unbounded: g' is not even bounded on $[-1, 1]$.

WORKED EXAMPLE 2 · $x^3\sin(1/x)$, FILL THE GAPS

$f(0)=0$. Quotient at 0: $x^2\sin\dfrac{1}{x}\to$?. For $x\neq 0$: $f'(x)=3x^2\sin\dfrac{1}{x}+x^3\cos\dfrac{1}{x}\cdot$? = ?. Is f' continuous at 0?

Answer: $f'(0)=0$; $f'(x)=3x^2\sin\dfrac{1}{x}-x\cos\dfrac{1}{x}$, and $|f'(x)|\le 3x^2+|x|\to 0=f'(0)$. Yes, continuous. So $r=2$ is the borderline: f' exists but is discontinuous; $r=3$ is smoother.

CHECK YOURSELF

$F(x)=x\sin(1/x)$, $F(0)=0$. Continuous at 0? Differentiable at 0?

Continuous: $|F(x)|\le|x|\to 0$. Not differentiable: the quotient is $\sin(1/x)$; $x_n=\dfrac{1}{n\pi}$ gives 0 and $y_n=\dfrac{2}{(4n+1)\pi}$ gives 1, so no limit.

YOUR INTUITION WILL LIE TO YOU HERE

"If f is differentiable everywhere, then f' is a nice continuous function." False (this example). Derivatives can be discontinuous, but (Darboux, later in this module) they can never have a **jump**.

In the model paper: **Q13** (Ex 6.1.12), **Q25**.

UNIT 2 · BARTLE §6.2

Mean values, extrema and monotonicity

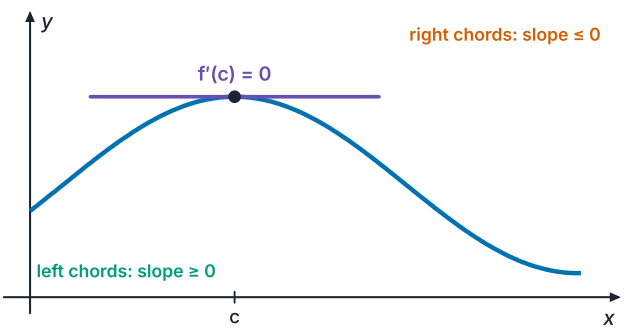
T10 Relative extrema; the Interior Extremum Theorem

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> Where can a hill-top or valley-bottom be? (related: CALC-24 Q10 critical points.)

PUZZLE

A smooth hill has a top. What must the slope be exactly at the top? Now ask the same of $f(x) = x$ on [0, 1], whose top is at x = 1. Same answer? Find out why not.



Interior extremum theorem. At an interior maximum, chords to the right slope down (≤ 0) and chords to the left slope up (≥ 0). If f′(c) exists it must be both ≥ 0 and ≤ 0, so f′(c) = 0.

DEFINITION · RELATIVE EXTREMUM (BARTLE §6.2)

Let $f : I \to \mathbb{R}$. f has a **relative maximum** at $c \in I$ if there is a neighbourhood $V = V_\delta(c)$ such that $f(x) \leq f(c)$ for all $x \in V \cap I$. f has a **relative minimum** at c if $f(c) \leq f(x)$ for all $x \in V \cap I$. A **relative extremum** is either.

"Relative" = best among the points near c only. Absolute = best on the whole domain.

THEOREM 6.2.1 · INTERIOR EXTREMUM THEOREM

Let c be an interior point of the interval I at which $f : I \to \mathbb{R}$ has a relative extremum. If the derivative of f at c exists, then $f'(c) = 0$.

Corollary 6.2.2: if f is continuous on I and has a relative extremum at an interior point c, then either $f'(c)$ does not exist or $f'(c) = 0$. These are the only places to look.

SKELETON · INTERIOR EXTREMUM (MAXIMUM CASE)

Let f have a relative maximum at the interior point c and let $f'(c)$ exist. Suppose $f'(c) > 0$. Then ? for x just right of c, so $f(x) > f(c)$: contradiction. Suppose $f'(c) < 0$. Then ? for x just left of c: contradiction. So $f'(c) = 0$. ∎ (Minimum: apply to $-f$.)

Know c interior; $f(x) \le f(c)$ for x near c; the quotient $\dfrac{f(x) - f(c)}{x - c}$ has a limit $f'(c)$.

Want $f'(c) = 0$. We show "> 0" and "< 0" are impossible.

PROOF (RELATIVE MAXIMUM)

1. Suppose $f'(c) > 0$.
   Proof by contradiction (M10): we kill both strict signs.

2. The quotient $\frac{f(x)-f(c)}{x-c}$ has limit $f'(c) > 0$, so there is a neighbourhood $V \subseteq I$ of c in which the quotient is > 0 for $x \neq c$.
   Local sign of a limit (Thm 4.2.9). "$V \subseteq I$" is possible because c is **interior**.

3. Take $x \in V$ with $x > c$. Then $f(x) - f(c) = (x-c) \cdot (\text{positive}) > 0$, i.e. $f(x) > f(c)$.
   Positive times positive.

4. Such x exist arbitrarily near c, which contradicts "$f(x) \leq f(c)$ near c".
   Definition of relative maximum.

5. Suppose $f'(c) < 0$. In the same V (shrunk), the quotient is < 0. Take $x < c$: $(x-c) < 0$ and quotient < 0 give $f(x) - f(c) > 0$. Contradiction again.
   Bartle writes "similarly"; this is the "how": the left side is where it breaks.

6. Hence $f'(c) = 0$. ∎
   Neither > 0 nor < 0 is possible.

HYPOTHESES CHECKLIST (WHAT FAILS WITHOUT EACH)

☐ c is **interior**

$f(x) = x$ on [0, 1]: max at 1, min at 0, but $f' = 1$ N7

☐ $f'(c)$ **exists**

$f(x) = |x|$ on [−1, 1]: min at 0, no derivative there C5

☐ (converse) $f'(c) = 0 \Rightarrow$ extremum?

x^3 at 0: $f'(0) = 0$ but no extremum C17

Where each was used. Interior: the neighbourhood V lies inside I, so points on both sides of c exist. Derivative exists: gives the limit whose sign we read. Extremum: forbids $f(x) > f(c)$ nearby.

YOUR INTUITION LIES · "CRITICAL POINT = EXTREMUM"

A critical point is only a **candidate**. x^3 at 0 has slope 0 and no extremum C17 . $|x|$ at 0 has no slope and is a minimum C5 . Endpoints are never interior, so check them separately.

WORKED EXAMPLE 1 · FULL (CALC-24 Q10)

Find all critical points of $f(x) = x^3 - 3x + 1$ and decide which are extrema.

$f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$, zero at $x = \pm 1$. f' exists everywhere, so any relative extremum is at ± 1 (Thm 6.2.1). Values: $f(-1) = -1 + 3 + 1 = 3$, $f(1) = 1 - 3 + 1 = -1$. Sign of f': $+$ on $(-\infty, -1)$, $-$ on $(-1, 1)$, $+$ on $(1, \infty)$, so a relative **max 3 at −1** and a relative **min −1 at 1** (First Derivative Test, T15).

On $[-2, 2]$ also test the ends: $f(-2) = -8 + 6 + 1 = -1$, $f(2) = 8 - 6 + 1 = 3$. Absolute max 3 (at −1 and at 2), absolute min −1 (at 1 and at −2).

WORKED EXAMPLE 2 · WITH GAPS

$f(x) = x^2 - 3x + 5$. $f'(x) =$? , zero at $x =$? . f at that point $=$? . Since f' is $-$ before and $+$ after, it is a relative ? . Answer: $f' = 2x - 3$, $x = \frac{3}{2}$, $f(\frac{3}{2}) = \frac{9}{4} - \frac{9}{2} + 5 = \frac{11}{4}$, a minimum.

CHECK YOURSELF

Find the absolute maximum and minimum of $g(x) = x^3 - 12x$ on $[-3, 3]$.

$g' = 3x^2 - 12 = 0$ at $x = \pm 2$. Compare $g(-3) = 9$, $g(-2) = 16$, $g(2) = -16$, $g(3) = -9$. Absolute max 16 at −2, absolute min −16 at 2.

In the model paper: **Q26** (state and prove 6.2.1), **Q14** (CALC-24 Q10).

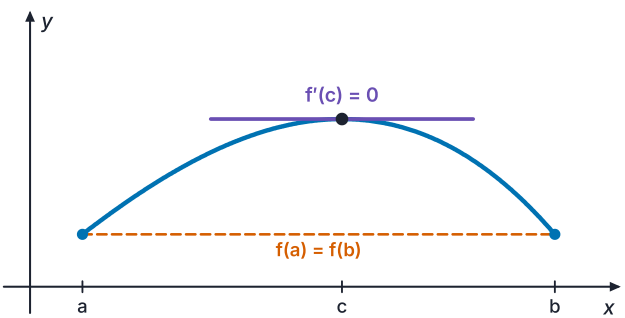
T11 Rolle's Theorem

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> Same height at both ends, so the tangent is flat somewhere between.

PUZZLE

You leave home and return home. Was there a moment when your height above sea level was not changing?



Rolle's theorem. f continuous on [a, b], differentiable on (a, b), f(a) = f(b). Then somewhere in between the tangent is horizontal: f′(c) = 0 for some c ∈ (a, b).

THEOREM 6.2.3 · ROLLE'S THEOREM

Suppose that f is continuous on a closed interval $I = [a, b]$, that the derivative f' exists at every point of the open interval (a, b), and that $f(a) = f(b) = 0$. Then there exists at least one point c in (a, b) such that $f'(c) = 0$.

"At least one": there may be several. If $f(a) = f(b) = h \neq 0$, apply the theorem to $f - h$.

SKELETON · ROLLE

If $f \equiv 0$, any c works. Otherwise f takes a positive (say) value. By the Max–Min Theorem f has a maximum at some $c \in [a, b]$ with $f(c) >$? . Then $c \neq a, b$ because ? . So c is interior and f has a relative maximum there, hence ? . ∎

Know f continuous on [a, b]; differentiable on (a, b); $f(a) = f(b) = 0$.

Want a point $c \in (a, b)$ with $f'(c) = 0$.

PROOF

1. If $f(x) = 0$ for all x, then $f' = 0$ and any $c \in (a, b)$ works.
   Trivial case first.

2. Otherwise f takes a nonzero value. Replacing f by $-f$ if necessary, assume f takes a positive value.
   $f'(c) = 0 \iff (-f)'(c) = 0$, so nothing is lost.

3. f is continuous on the closed bounded interval [a, b], so it attains a maximum $M = f(c)$ at some $c \in [a, b]$; and $M > 0$.
   Max–Min Theorem 5.3.4 (Module 1, T9). Needs closed, bounded, continuous.

4. $c \neq a$ and $c \neq b$, since $f(a) = f(b) = 0 < M$. So $c \in (a, b)$.
   This is why the equal end values are in the hypotheses.

5. f is differentiable at the interior point c and has a relative (indeed absolute) maximum there, so $f'(c) = 0$. ∎
   Interior Extremum Theorem 6.2.1 (T10). Differentiability is needed only on the open interval.

HYPOTHESES CHECKLIST

☐ f continuous on the **closed** [a, b]
$f(x) = x$ on $[0, 1)$, $f(1) = 0$: differentiable inside, ends both 0, but $f' = 1$ **C18**

☐ f differentiable on (a, b)
$f(x) = 1 - |x|$ on $[-1, 1]$: ends 0, slope only ± 1 **C18**

☐ $f(a) = f(b)$
$f(x) = x$ on $[0, 1]$: $f' = 1$ **C18**

Where each was used. Continuity on [a, b]: Max–Min gives the maximum point. Equal ends: the maximum is not at an end. Differentiability on (a, b): $f'(c)$ exists at the interior maximum, so 6.2.1 applies.

YOUR INTUITION LIES · "C IS THE TOP"

Rolle gives *some* flat point, not necessarily the highest point, and not only one. It may be a hill-top, a valley or a flat shoulder.

WORKED EXAMPLE 1 · FULL

Verify Rolle's Theorem for $f(x) = x(x-1)(x-2)$ on $[0, 2]$.

f is a polynomial: continuous and differentiable everywhere. $f(0) = 0 = f(2)$. $f(x) = x^3 - 3x^2 + 2x$, so $f'(x) = 3x^2 - 6x + 2$. $f'(c) = 0$ gives $c = \dfrac{6 \pm \sqrt{36 - 24}}{6} = 1 \pm \dfrac{1}{\sqrt{3}}$, that is $c \approx 0.4226$ and $c \approx 1.5774$. Both lie in $(0, 2)$. Values $f(c) = \pm\dfrac{2}{3\sqrt{3}} \approx \pm 0.3849$. Two points, not one.

WORKED EXAMPLE 2 · WITH GAPS

$f(x) = x^3 - 3x$ on $[0, \sqrt{3}]$. $f(0) =$? and $f(\sqrt{3}) =$? . $f'(x) =$? $= 0$ at $c =$? $\in (0, \sqrt{3})$. Answer: 0 and $3\sqrt{3} - 3\sqrt{3} = 0$; $f' = 3x^2 - 3$; $c = 1$ (reject −1, outside the interval).

CHECK YOURSELF

Does Rolle's Theorem apply to $f(x) = 1 - x^{2/3}$ on $[-1, 1]$? Is there a c with $f'(c) = 0$?

$f(\pm 1) = 0$ and f is continuous, but $f'(x) = -\frac{2}{3}x^{-1/3}$ does not exist at 0 (vertical tangent, a cusp). So the hypothesis "differentiable on (−1, 1)" fails, and indeed $f' \neq 0$ everywhere it exists. The theorem is silent here.

In the model paper: **Q27** (state, prove, verify for $x(x-1)(x-2)$).

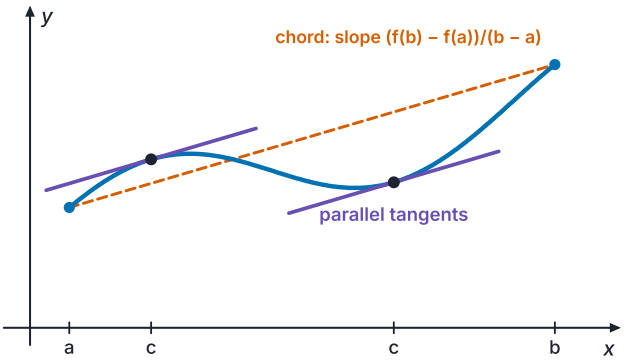
T12 The Mean Value Theorem

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> The most useful theorem of the module: Rolle, tilted.

PUZZLE

A car covers 120 km in exactly 2 hours. Must the speedometer have read exactly 60 km/h at some instant? (Yes. Why?)



Mean value theorem: tilted Rolle. Somewhere the tangent is parallel to the chord: f(b) − f(a) = f′(c)(b − a). Here there are two such c. The theorem promises at least one, and says nothing about where.

THEOREM 6.2.4 · MEAN VALUE THEOREM

Suppose that f is continuous on a closed interval $I = [a, b]$, and that f has a derivative in the open interval (a, b). Then there exists at least one point c in (a, b) such that

$$f(b) - f(a) = f'(c)(b - a).$$

Reading: the average slope over [a, b] equals the instantaneous slope at some c. Geometrically: some tangent is parallel to the chord.

SKELETON · MVT

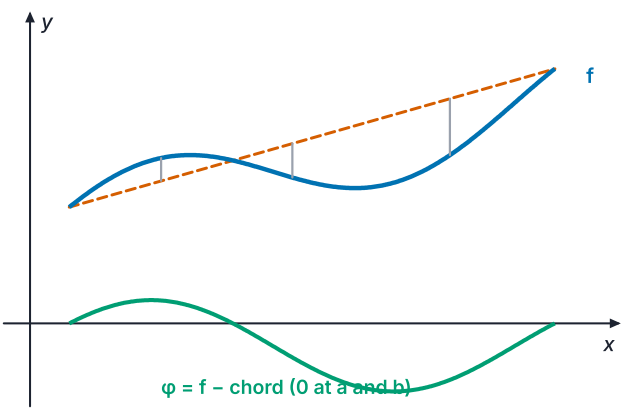
Define $\varphi(x) = f(x) - f(a) -$? $\cdot(x - a)$ (f minus the chord). Then φ is continuous on [a, b], differentiable on (a, b), and $\varphi(a) =$?, $\varphi(b) =$?. By Rolle there is c with $\varphi'(c) = 0$, which says ? . ∎

SCRATCH · HOW TO FIND Φ (NEVER COPIED INTO THE EXAM)

Want: a function that is 0 at both ends, so Rolle applies.

The chord is the line through $(a, f(a))$ and $(b, f(b))$: $y = f(a) + m(x - a)$ with $m = \dfrac{f(b) - f(a)}{b - a}$.

Take the **gap** $\varphi = f - \text{chord}$: it is 0 at a and at b automatically. Differentiate: $\varphi' = f' - m$, and $\varphi' = 0$ is exactly "$f' = m$".



Proof idea of the MVT. φ(x) = f(x) − [chord] is the vertical gap (grey) between the graph and the chord. It is 0 at both ends, so Rolle gives φ′(c) = 0, i.e. f′(c) = slope of the chord.

PROOF (BARTLE)

1. Let $\varphi(x) = f(x) - f(a) - \dfrac{f(b) - f(a)}{b - a}(x - a)$ for $x \in [a, b]$.
   f minus the chord: the vertical gap in the picture.

2. φ is continuous on [a, b] and differentiable on (a, b).
   f is, and the subtracted term is a straight line.

3. $\varphi(a) = 0$, and $\varphi(b) = f(b) - f(a) - (f(b) - f(a)) = 0$.
   Substitute $x = a$ and $x = b$.

4. By Rolle there is $c \in (a, b)$ with $\varphi'(c) = 0$.
   Theorem 6.2.3 applied to φ.

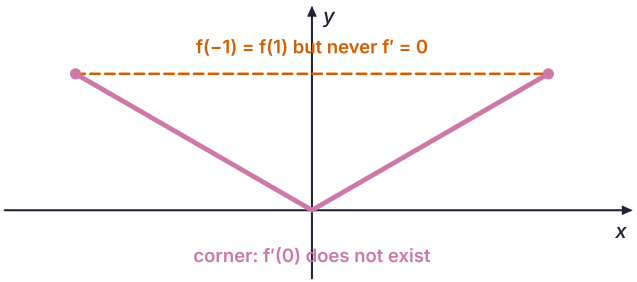
5. $0 = \varphi'(c) = f'(c) - \dfrac{f(b) - f(a)}{b - a}$, so $f(b) - f(a) = f'(c)(b - a)$. ∎
   Rearrange.

HYPOTHESES CHECKLIST

☐ f continuous on [a, b], *including the ends*
$f(x) = x$ on $[0, 1)$, $f(1) = 0$: chord slope 0, $f' = 1$ **C18**

☐ f differentiable on (a, b)
$|x|$ on $[-1, 1]$: chord slope 0, $f' = \pm 1$ **C18**

Where each was used. Both only enter through Rolle for φ : continuity on [a, b] makes φ continuous on [a, b] (Max–Min); differentiability on (a, b) makes $\varphi'(c)$ exist at the interior maximum.



Why "differentiable on (a, b)" is needed. $|x|$ on $[-1, 1]$ has equal end values, but its slope is only ±1, never 0 C18.

YOUR INTUITION LIES · "C IS THE MIDPOINT"

c is some point in (a, b); it is not unique and it is not the midpoint. Only for a quadratic is it the midpoint (x^2 on [1, 3]: $2c = 4$, $c = 2$). For x^3 on [0, 2] below, $c \approx 1.155 \neq 1$ N12.

WORKED EXAMPLE 1 · FULL

Verify the MVT for $f(x) = x^3$ on $[0, 2]$ and find c.

f is a polynomial. Chord slope $\frac{f(2) - f(0)}{2 - 0} = \frac{8}{2} = 4$. Solve $f'(c) = 3c^2 = 4$: $c = \pm\frac{2}{\sqrt{3}} \approx \pm 1.1547$. Only $c = \frac{2}{\sqrt{3}} \approx 1.1547$ lies in $(0, 2)$. Check: $f(c) \approx 1.5396$ and tangent slope 4 there, parallel to the chord $y = 4x$.

WORKED EXAMPLE 2 · WITH GAPS

$f(x) = \sqrt{x}$ on $[1, 4]$. Chord slope $= \dfrac{f(4) - f(1)}{4 - 1} =$? . $f'(c) = \dfrac{1}{2\sqrt{c}} =$? gives $\sqrt{c} =$? , so $c =$? $\in (1, 4)$. Answer: $\dfrac{2-1}{3} = \dfrac{1}{3}$; $\dfrac{1}{2\sqrt{c}} = \dfrac{1}{3}$; $\sqrt{c} = \dfrac{3}{2}$; $c = \dfrac{9}{4}$.

CHECK YOURSELF

Find the number c promised by the MVT for $f(x) = \ln x$ on $[1, e]$.

Chord slope $\dfrac{\ln e - \ln 1}{e - 1} = \dfrac{1}{e-1}$. $f'(c) = \dfrac{1}{c} = \dfrac{1}{e-1}$, so $c = e - 1 \approx 1.718 \in (1, e)$.

In the model paper: **Q38** (state and prove, 10 marks), **Q16** (verify for x^3).

T13 Zero derivative means constant

index ↑

> If the speed is 0 all the time, you did not move.

PUZZLE

$\sin^2 x + \cos^2 x$ has derivative $2 \sin x \cos x - 2 \cos x \sin x = 0$. What does that tell you without any trigonometry?

THEOREM 6.2.5

If f is continuous on the closed interval $I = [a, b]$, differentiable on (a, b), and $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on I.

Picture: a graph with horizontal tangent everywhere cannot go up or down.

COROLLARY 6.2.6

If f, g are continuous on $[a, b]$, differentiable on (a, b) and $f'(x) = g'(x)$ for all $x \in (a, b)$, then there is a constant C with $f = g + C$ on $[a, b]$.

SKELETON

Fix $x \in (a, b]$. Apply the MVT to f on ? . Then $f(x) - f(a) = f'(c)(x - a) =$? . So $f(x) = f(a)$ for every x: constant. For the corollary apply this to $h =$? . ∎

PROOF

1. Let $x \in (a, b]$. f is continuous on $[a, x]$ and differentiable on (a, x).
   Restrict the hypotheses of [a, b] to the smaller closed interval.

2. By the MVT, $f(x) - f(a) = f'(c)(x - a)$ for some $c \in (a, x) \subseteq (a, b)$.
   Theorem 6.2.4 (T12).

3. $f'(c) = 0$, so $f(x) = f(a)$. This holds for every x: f is constant. ∎
   c lies in (a, b) where $f' = 0$.

4. Corollary: put $h = f - g$. Then $h' = f' - g' = 0$ on (a, b), so $h \equiv C$.
   Apply the theorem to h.

HYPOTHESES CHECKLIST

☐ domain is one **interval**

$f = 0$ on $(0,1)$, $f = 1$ on $(2,3)$: $f' = 0$ but not constant N5

☐ f continuous at the ends

$f(0) = 5$, $f(x) = 0$ for $0 < x \leq 1$: $f' = 0$ on (0, 1) but not constant

Where each was used. Interval: MVT needs $[a, x] \subseteq I$. Continuity at the ends: MVT is used on $[a, x]$, which includes a.

YOUR INTUITION LIES · TWO PIECES

$g(x) = |x|/x$ has $g'(x) = 0$ for every $x \neq 0$, yet $g = -1$ on $(-\infty, 0)$ and $+1$ on $(0, \infty)$. "Constant" is only claimed on an interval N5. Likewise $F' = 1/x$ on $\mathbb{R} \setminus \{0\}$ gives $\ln|x| + C_1$ on one side and $\ln|x| + C_2$ on the other.

WORKED EXAMPLE 1 · FULL

Show $\sin^2 x + \cos^2 x = 1$ for all x using only derivatives.

Let $h(x) = \sin^2 x + \cos^2 x$. $h'(x) = 2\sin x \cos x + 2\cos x(-\sin x) = 0$ for all x ($\sin' = \cos$, $\cos' = -\sin$, Plus Two facts). h is differentiable, hence continuous, on every $[0, x]$ (or $[x, 0]$). By Thm 6.2.5, $h(x) = h(0) = 0 + 1 = 1$.

WORKED EXAMPLE 2 · WITH GAPS

If $f'(x) = 2x$ for all $x \in \mathbb{R}$ and $f(0) = 3$, find f.
Let $g(x) = x^2$; $g' =$? $= f'$, so $f - g =$? .
At $x = 0$: $C =$? . Hence $f(x) =$? .
Answer: $2x$; a constant C; $C = 3 - 0 = 3$;
$f(x) = x^2 + 3$.

CHECK YOURSELF

$g(x) = |x|/x$ has $g' = 0$ at every point where it is defined. Why is this not a contradiction with Theorem 6.2.5?

Its domain $\mathbb{R} \setminus \{0\}$ is not an interval, so the MVT cannot join the two pieces (there is no point 0 to differentiate through).

In the model paper: **Q28** (prove, deduce $f = g + C$, explain failure on $(0,1) \cup (2,3)$).

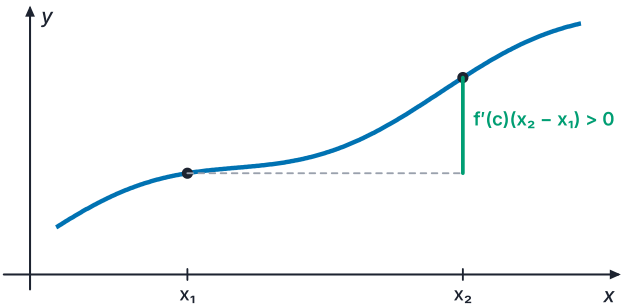
T14 Increasing and decreasing: the sign of f′

index ↑

Intervals of increase and decrease (CALC-24 Q9, Q20(1); old MCQ MTS-22 MCQ 5).

PUZZLE

x^3 is always climbing, yet its derivative is 0 at x = 0. And $f'(0) > 0$ at a single point does not make a function climb near that point. Both are true: see below.



f′ > 0 on an interval ⇒ f is increasing there. MVT: $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$, and both factors are positive.

DEFINITION · INCREASING, DECREASING (BARTLE)

$f : I \to \mathbb{R}$ is **increasing** on I if $x_1, x_2 \in I$, $x_1 < x_2$ imply $f(x_1) \leq f(x_2)$; **strictly increasing** if $f(x_1) < f(x_2)$. **Decreasing**: $f(x_1) \geq f(x_2)$; strictly decreasing: $>$.

Bartle's "increasing" is non-strict, so a constant function is both increasing and decreasing N14.

THEOREM 6.2.7 · MONOTONICITY

Let $f : I \to \mathbb{R}$ be differentiable on the interval I. Then (a) f is increasing on I if and only if $f'(x) \geq 0$ for all $x \in I$; (b) f is decreasing on I if and only if $f'(x) \leq 0$ for all $x \in I$.

Strict version: if $f' > 0$ on I then f is strictly increasing. The converse is false (x^3).

SKELETON · (A), TWO DIRECTIONS

(⇐) Let $f' \geq 0$ on I and $x_1 < x_2$ in I. By the ? on $[x_1, x_2]$, $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) \geq$? . (⇒) Let f be increasing and $c \in I$. For $x \neq c$ the quotient $\dfrac{f(x) - f(c)}{x - c}$ is ? , so its limit $f'(c)$ is ? . ∎

PROOF OF (A) (CORRECT LABELLING: BARTLE'S TEXT SWAPS THE ARROWS)

1. **(⇐)** Suppose $f'(x) \geq 0$ on I. Let $x_1 < x_2$ in I.
   We prove "f' ≥ 0 gives increasing".

2. f is differentiable on I, hence continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2). The MVT gives $c \in (x_1, x_2)$ with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$.
   Differentiable ⇒ continuous (Thm 6.1.2); MVT 6.2.4.

3. $f'(c) \geq 0$ and $x_2 - x_1 > 0$, so $f(x_2) - f(x_1) \geq 0$.
   Sign of a product. This is the step that is strict if $f' > 0$.

4. **(⇒)** Suppose f is increasing and $c \in I$. For $x \in I$, $x \neq c$: numerator and denominator of $\dfrac{f(x) - f(c)}{x - c}$ have the same sign (or the numerator is 0), so the quotient is ≥ 0.
   Definition of increasing, both sides of c.

5. Let $x \to c$: $f'(c) \geq 0$. ∎
   Order Theorem 4.2.6: "≥ 0" survives a limit (it would not stay "> 0").

Part (b): apply (a) to $-f$.

HYPOTHESES CHECKLIST

☐ f differentiable at **every** point of I

$f = 1$ for $x < 0$, $f = 0$ for $x \geq 0$: $f' = 0 \geq 0$ wherever it exists, yet f decreases (no derivative at 0)

☐ I is one **interval**

$f(x) = -1/x$ on $\mathbb{R} \setminus \{0\}$: $f' = 1/x^2 > 0$, but $f(-1) = 1 > -1 = f(1)$ N6

☐ (strict) $f' > 0$ is only *sufficient*

x^3 is strictly increasing with $f'(0) = 0$ C17

Where each was used. Interval: the MVT needs $[x_1, x_2] \subseteq I$. Differentiable at every point: continuity for the MVT and a quotient limit for (⇒).

YOUR INTUITION LIES · "F′(C) > 0 ⇒ F INCREASES NEAR C"

False. $g(x) = x + 2x^2 \sin(1/x)$, $g(0) = 0$. Then $g'(0) = \lim (1 + 2x \sin \frac{1}{x}) = 1 > 0$. For $x \neq 0$, $g'(x) = 1 + 4x \sin \frac{1}{x} - 2 \cos \frac{1}{x}$. At $x = \frac{1}{2n\pi}$: $g' = 1 + 0 - 2 = -1 < 0$. At $x = \frac{1}{(2n+1)\pi}$: $g' = 1 + 0 + 2 = 3 > 0$. Both kinds of points crowd into 0, so g is not monotone in any neighbourhood of 0. (Bartle Ex 6.2.10.) The local sign lemma (T22) only compares $g(x)$ with $g(0)$.

METHOD M05 · SIGN ANALYSIS OF F′

1. Compute f' and factorise. 2. Solve $f' = 0$ (and note where f' is undefined). 3. Mark these points on a number line. 4. Test one point in each gap, write + or −. 5. Write the intervals **including endpoints** (Bartle's definition is non-strict).

WORKED EXAMPLE 1 · FULL (CALC-24 Q20(1))

Where is $f(x) = x^4 - 5x^3 + 9x^2$ increasing and decreasing?

$f'(x) = 4x^3 - 15x^2 + 18x = x(4x^2 - 15x + 18)$. The quadratic has discriminant $15^2 - 4 \cdot 4 \cdot 18 = 225 - 288 = -63 < 0$ and leading coefficient $4 > 0$, so it is always positive. Therefore f' has the sign of x: $f' \leq 0$ on $(-\infty, 0]$, $f' \geq 0$ on $[0, \infty)$. **Decreasing on** $(-\infty, 0]$**, increasing on** $[0, \infty)$; minimum $f(0) = 0$.

WORKED EXAMPLE 2 · WITH GAPS

$f(x) = 3x - 4x^2$. $f'(x) =$? $= 0$ at $x =$?. $f' > 0$ for $x <$?. So f is increasing on ? and decreasing on ?; maximum value ?. Answer: $3 - 8x$; $\frac{3}{8}$; $\frac{3}{8}$; $(-\infty, \frac{3}{8}]$; $[\frac{3}{8}, \infty)$; $f(\frac{3}{8}) = \frac{9}{8} - \frac{9}{16} = \frac{9}{16}$.

CHECK YOURSELF

Find the intervals of increase and decrease of $f(x) = 2x^3 - 9x^2 + 12x$.

$f' = 6x^2 - 18x + 12 = 6(x-1)(x-2)$. Increasing on $(-\infty, 1]$, decreasing on $[1, 2]$, increasing on $[2, \infty)$. (Values: $f(1) = 5$ local max, $f(2) = 4$ local min.)

OLD MCQ · ADAPTED

Which of $1 - x^3$, $x^2 + x$, $1/x^2$ increase on $(0, 1)$? Derivatives $-3x^2 < 0$, $2x + 1 > 0$, $-2/x^3 < 0$ on (0, 1). Only $x^2 + x$.

In the model paper: **Q40** (prove and apply to $x - \sin x$), **Q17** (CALC-24 Q9), **Q18** (CALC-24 Q20(1)), **Q19** (MTS-22 MCQ 5, adapted).

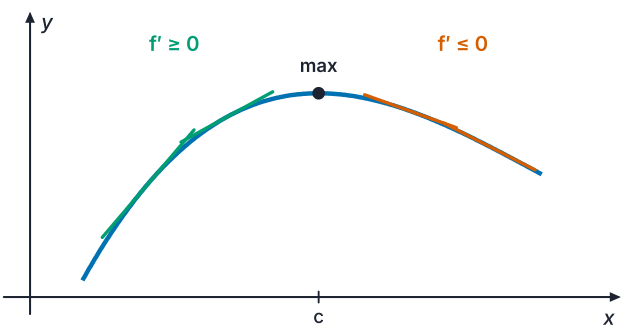
T15 The First Derivative Test

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> Sign change of f′ means an extremum (CALC-24 Q18).

PUZZLE

$|x|$ has a minimum at 0 but no derivative there. What does the test need instead of $f'(0)$?



First derivative test. f′ ≥ 0 just left of c and f′ ≤ 0 just right of c ⇒ f has a relative maximum at c. (Reverse signs: minimum.) f need not be differentiable at c itself, only continuous.

THEOREM 6.2.8 · FIRST DERIVATIVE TEST

Let f be continuous on the interval $I = [a, b]$ and let c be an interior point of I. Assume that f is differentiable on (a, c) and (c, b).
(a) If there is a neighbourhood $(c - \delta, c + \delta) \subseteq I$ such that $f'(x) \geq 0$ for $c - \delta < x < c$ and $f'(x) \leq 0$ for $c < x < c + \delta$, then f has a relative maximum at c.

(b) If $f'(x) \leq 0$ for $c - \delta < x < c$ and $f'(x) \geq 0$ for $c < x < c + \delta$, then f has a relative minimum at c.

Up then down = hill-top; down then up = valley. $f'(c)$ itself need not exist.

SKELETON · (A)

Left of c: for $x \in (c - \delta, c)$ apply the MVT on ? to get $f(c) - f(x) = f'(\xi)(c - x) \geq 0$.
Right of c: for $x \in (c, c + \delta)$ apply MVT on ? to get $f(x) - f(c) = f'(\xi)(x - c) \leq 0$. So $f(x) \leq f(c)$ on the whole neighbourhood. ∎

PROOF OF (A)

1. Let $x \in (c-\delta, c)$. f is continuous on $[x, c]$ and differentiable on (x, c).
   Differentiable on (a, c), and continuous at c by hypothesis.

2. MVT: $f(c) - f(x) = f'(\xi)(c - x)$ for some $\xi \in (x, c)$. Here $f'(\xi) \geq 0$ and $c - x > 0$, so $f(c) \geq f(x)$.
   ξ lies in $(c-\delta, c)$ where $f' \geq 0$.

3. Let $x \in (c, c+\delta)$. MVT on $[c, x]$: $f(x) - f(c) = f'(\xi)(x - c)$, $\xi \in (c, x)$, and $f'(\xi) \leq 0$, so $f(x) \leq f(c)$.
   Bartle says "how?"; this is how. Now ξ lies in $(c, c+\delta)$.

4. So $f(x) \leq f(c)$ for all x in $(c-\delta, c+\delta)$: relative maximum at c. ∎
   Definition of relative maximum. (b): apply (a) to $-f$.

HYPOTHESES CHECKLIST

☐ f **continuous at c**

$f(x) = -x^2$ for $x \neq 0$, $f(0) = -1$: $f' \geq 0$ left, ≤ 0 right, but 0 is a strict *minimum*

☐ sign of f′ on **both** sides

$f' \geq 0$ on both sides: x^3 at 0, no extremum C17

☐ (converse) extremum ⇒ sign change?

$f(x) = 2x^4 + x^4 \sin(1/x)$, $f(0) = 0$: minimum at 0 but f' changes sign infinitely often near 0 N9

Where each was used. Continuity at c: lets the MVT use the closed interval $[x, c]$. Signs of f′ on each side: bound the MVT slope. Differentiability off c: the MVT's open interval.

YOUR INTUITION LIES · "AN EXTREMUM MUST HAVE F′ CHANGING SIGN"

For $f(x) = 2x^4 + x^4 \sin(1/x)$: $f \geq x^4 \geq 0 = f(0)$, so 0 is an absolute minimum, and $f'(0) = 0$. But $f'(x) = 8x^3 + 4x^3 \sin\dfrac{1}{x} - x^2 \cos\dfrac{1}{x}$ equals $x^2(8x - 1) < 0$ at $x = \dfrac{1}{2n\pi}$ and $x^2(8x + 1) > 0$ at $x = \dfrac{1}{(2n+1)\pi}$ (and likewise for negative x). The sign of f′ flips arbitrarily close to 0 on both sides. (Bartle Ex 6.2.9.)

WORKED EXAMPLE 1 · FULL

Relative extrema of $f(x) = x^3 - 3x$ (and of $|x|$ at 0).

$f'(x) = 3(x^2 - 1) = 3(x-1)(x+1)$. For $x < -1$: $f' > 0$; for $-1 < x < 1$: $f' < 0$; for $x > 1$: $f' > 0$. At −1 the sign goes + to −: relative **max** $f(-1) = -1 + 3 = 2$. At 1 it goes − to +: relative **min** $f(1) = 1 - 3 = -2$. For $|x|$: $f' = -1$ for $x < 0$, $+1$ for $x > 0$: minimum at 0, although $f'(0)$ does not exist.

WORKED EXAMPLE 2 · WITH GAPS (CALC-24 Q18)

$f(x) = \dfrac{x^2}{x^4 + 16}$. $f'(x) = \dfrac{2x(16 - x^4)}{(x^4 + 16)^2}$. Zeros: ? . Sign of f' on $(-\infty, -2)$, $(-2, 0)$, $(0, 2)$, $(2, \infty)$: ? . So maxima at ? with value ? and a minimum at ? with value ? .

Answer: zeros $0, \pm 2$. Signs $+, -, +, -$. Maxima at $x = \pm 2$, $f(\pm 2) = \dfrac{4}{32} = \dfrac{1}{8}$; minimum at 0, $f(0) = 0$.

CHECK YOURSELF

Find the relative extrema of $f(x) = x^2 e^{-x}$.

$f'(x) = (2x - x^2)e^{-x} = x(2 - x)e^{-x}$. Signs: $-$ on $(-\infty, 0)$, $+$ on $(0, 2)$, $-$ on $(2, \infty)$. Minimum at 0 ($f = 0$), maximum at 2 ($f = 4e^{-2} \approx 0.541$).

In the model paper: **Q29** (prove (a), apply to $x^3 - 3x$ and $|x|$), **Q15** (CALC-24 Q18).

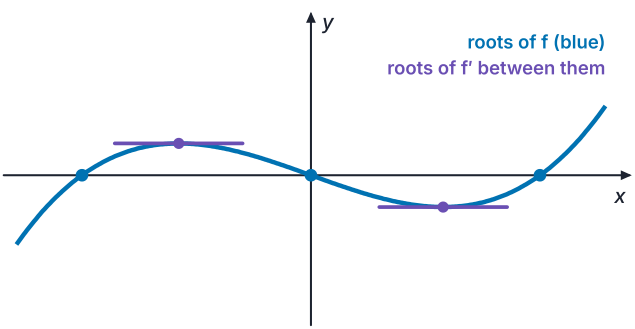
T16 Rolle for roots: location, interlacing, counting

index ↑

> "Show the equation has exactly one real root" (MAT-21 Q17 style).

PUZZLE

The roots of $\sin x$ are $0, \pm\pi, \pm 2\pi, \ldots$ and the roots of $\cos x$ are $\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$ They alternate. Is that an accident?



Counting roots with Rolle. Between two roots of f there is a root of f′. So if f′ has at most k roots, f has at most k + 1.

CONSEQUENCE OF ROLLE'S THEOREM (BARTLE 6.2.9(A))

Let f be continuous on $[r_1, r_2]$ and differentiable on (r_1, r_2). If $f(r_1) = f(r_2) = 0$ (two roots), then f' has a root in (r_1, r_2).

Equivalently: if $f' \neq 0$ on an interval, f has **at most one** root there. If f' has at most k roots there, f has at most $k + 1$.

SKELETON · "EXACTLY ONE ROOT"

Existence: f is continuous; $f(a) =$? < 0 and $f(b) =$? > 0, so by the Location of Roots Theorem (Module 1) f has a root in (a, b).
Uniqueness: suppose f had two roots $r_1 < r_2$. By Rolle $f'(c) = 0$ for some c between them. But $f'(x) =$? $\neq 0$ for all x. Contradiction. ∎

HYPOTHESES CHECKLIST

☐ f differentiable *between* the two roots
$f(x) = |x| - 1$ has roots ±1, but $f' = \pm 1 \neq 0$: not differentiable at 0 C18

☐ (uniqueness part) $f' \neq 0$ on the **whole** interval
$x^3 - x$ has $f'(\pm\frac{1}{\sqrt{3}}) = 0$ and three roots

YOUR INTUITION LIES · "F' ≠ 0 MEANS NO ROOT"

$f' \neq 0$ limits f to *at most* one root. To show it has one, you still need a sign change (Location of Roots, T10 of Module 1).

WORKED EXAMPLE 1 · FULL

Show $p(x) = x^3 + 3x + 1$ has exactly one real root and locate it.

Existence. $p(-1) = -1 - 3 + 1 = -3 < 0$ and $p(0) = 1 > 0$; p is continuous, so a root lies in $(-1, 0)$. **Uniqueness.** $p'(x) = 3x^2 + 3 \geq 3 > 0$ for all x. Two roots would give c with $p'(c) = 0$ by Rolle: impossible. **Locating.** $p(-0.5) = -0.125 - 1.5 + 1 = -0.625 < 0$ and $p(-0.25) = -0.015625 - 0.75 + 1 = 0.234375 > 0$, so the root is in $(-0.5, -0.25)$ (it is about -0.3222).

WORKED EXAMPLE 2 · WITH GAPS (MAT-21 Q17)

$f(x) = xe^x - 2$. $f(0) =$? , $f(1) =$? > 0, so a root in (0, 1). $f'(x) =$? > 0 on (0, 1), so there is ? root in (0, 1). Answer: $f(0) = -2$; $f(1) = e - 2 \approx 0.718$; $f' = (x + 1)e^x$; exactly one (it is about 0.8526).

CHECK YOURSELF

Show that $x^5 + x - 1 = 0$ has exactly one real root, and say in which interval of length 1 it lies.

$f(0) = -1 < 0 < 1 = f(1)$: a root in (0, 1). $f'(x) = 5x^4 + 1 \geq 1 > 0$, so by Rolle at most one root. Exactly one, in (0, 1).

INTERLACING

$\sin' = \cos$: between two roots of sin there is a root of cos. $\cos' = -\sin$: between two roots of cos there is a root of sin. So the roots alternate. (Plus Two facts for $\sin'$ and $\cos'$.)

In the model paper: **Q30** (interlacing and exactly one root of $x^3 + 3x + 1$).

T17 MVT for approximate calculation

index ↑

> Trap a number between two decimals (Bartle 6.2.9(b)).

PUZZLE

Without a calculator, how close can you pin down $\sqrt{105}$ knowing only $\sqrt{100} = 10$ and $\sqrt{121} = 11$?

METHOD M06 · BOUNDS FROM THE MVT

To estimate $F(b)$ from the known $F(a)$: apply the MVT to F on $[a, b]$, so $F(b) - F(a) = F'(c)(b - a)$ for some $c \in (a, b)$; then bound $F'(c)$ using $a < c < b$.

SKELETON

Let $f(x) =$? on $[a, b] =$? . f is continuous on [a, b] and differentiable on (a, b). By the MVT, $f(b) - f(a) = f'(c)(b - a)$ with $a < c < b$. Since f' is ? , $f'(c)$ lies between ? and ? . Multiply by $b - a > 0$ (direction kept). ∎

WORKED EXAMPLE 1 · FULL (BARTLE 6.2.9(B))

Bound $\sqrt{105}$.

Let $f(x) = \sqrt{x}$ on $[100, 105]$; $f'(x) = \dfrac{1}{2\sqrt{x}}$. MVT: $\sqrt{105} - 10 = \dfrac{5}{2\sqrt{c}}$ with $100 < c < 105$.

First pass. $10 < \sqrt{c} < \sqrt{105} < \sqrt{121} = 11$, so $\dfrac{5}{22} < \sqrt{105} - 10 < \dfrac{5}{20}$, i.e. $10.2272 < \sqrt{105} < 10.25$.

Sharpen. Since $10.25^2 = 105.0625 > 105$, we know $\sqrt{105} < 10.25$, so $\sqrt{c} < 10.25$ and $\dfrac{5}{2\sqrt{c}} > \dfrac{5}{20.5} = 0.2439$. Therefore $10.2439 < \sqrt{105} < 10.25$. (True value 10.24695.)

WORKED EXAMPLE 2 · WITH GAPS

Bound $\sqrt[3]{28}$. $f(x) = x^{1/3}$ on $[27, 28]$, $f' = \dfrac{1}{3}x^{-2/3}$. $\sqrt[3]{28} - 3 = \dfrac{1}{3c^{2/3}}$, $27 < c < 28$. Then $c^{2/3} > 27^{2/3} =$? , so $\sqrt[3]{28} - 3 <$? . Also $c^{2/3} < 28^{2/3} < 3.1^2 =$? (as $3.1^3 = 29.791 > 28$), so $\sqrt[3]{28} - 3 >$? . Answer: 9; $\dfrac{1}{27} \approx 0.0370$; 9.61; $\dfrac{1}{28.83} \approx 0.0347$. So $3.0346 < \sqrt[3]{28} < 3.0371$ (true value 3.03659).

YOUR INTUITION LIES · "C IS KNOWN"

The MVT does not tell you c, only that $a < c < b$. All the work is in a **bound** on f' over the whole interval N12 . Using $c \approx b$ or $c \approx a$ gives a one-sided answer only if you say which.

CHECK YOURSELF

Use the MVT for $\ln$ on $[1, 1.1]$ to show $0.0909 < \ln 1.1 < 0.1$.

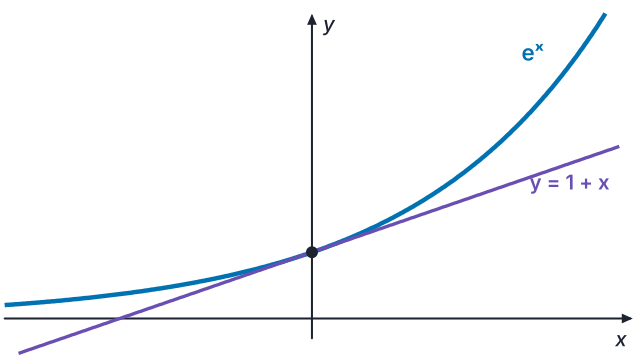
$\ln 1.1 - \ln 1 = \frac{1}{c}(0.1)$, $1 < c < 1.1$. So $\frac{1}{1.1} < \frac{1}{c} < 1$ and $\frac{0.1}{1.1} = 0.0909\ldots < \ln 1.1 < 0.1$. (True value 0.09531.)

In the model paper: **Q31** ($\sqrt{105}$ and sharpening).

T18 Inequalities from the MVT: exponential and logarithm

index ↑

$e^x \geq 1 + x$ and $\frac{x-1}{x} < \ln x < x - 1$ (Bartle 6.2.10(a), Ex 6.2.6, 6.2.7).



$e^x \geq 1 + x$ for all x, with equality only at 0. The line 1 + x is the tangent at 0, and the curve always lies above it. Proof: MVT on [0, x] (or [x, 0]).

RESULTS (BARTLE 6.2.10(A) AND EX 6.2.7)

(a) $e^x \geq 1 + x$ for all $x \in \mathbb{R}$, with equality only at $x = 0$. (b) For $x > 1$: $\frac{x-1}{x} < \ln x < x - 1$.

(Plus Two facts: $(e^x)' = e^x$, $(\ln x)' = \frac{1}{x}$, $e^x > 1$ for $x > 0$, $e^x < 1$ for $x < 0$.)

SKELETON · TEMPLATE FOR EVERY MVT INEQUALITY

1. Name f and the interval. 2. State: f continuous on closed, differentiable on open. 3. Write $f(b) - f(a) = f'(c)(b - a)$. 4. Bound $f'(c)$ using the position of c. 5. Multiply by $b - a > 0$; strict if the bound is strict. 6. Treat $x = a$ and the other side separately.

SCRATCH · WHICH FUNCTION, WHICH INTERVAL?

Want $e^x \geq 1 + x$, i.e. $e^x - e^0 \geq x - 0$.
Take $f = \exp$ on $[0, x]$ (or $[x, 0]$). Then $e^x - 1 = e^c \cdot x$.
The size of e^c versus 1 depends on the sign of c, so split $x > 0$ and $x < 0$.

PROOF OF (A)

1. **x = 0:** both sides equal 1.
   Equality case.

2. **x > 0:** MVT for e^t on $[0, x]$: $e^x - 1 = e^c\, x$ with $0 < c < x$. Since $c > 0$, $e^c > 1$, so $e^x - 1 > x$.
   $x > 0$, so multiplying $e^c > 1$ by x keeps ">".

3. **x < 0:** MVT on $[x, 0]$: $1 - e^x = e^c(0 - x) = (-x)e^c$ with $x < c < 0$. Since $c < 0$, $e^c < 1$ and $-x > 0$, so $1 - e^x < -x$.
   Multiply $e^c < 1$ by the positive number $-x$.

4. Rearranging: $e^x > 1 + x$. ∎
   Equality only at $x = 0$.

PROOF OF (B)

1. Let $x > 1$ and apply the MVT to $\ln$ on $[1, x]$: $\ln x - \ln 1 = \frac{1}{c}(x - 1)$ for some $c \in (1, x)$.
   $\ln$ is continuous on [1, x] and differentiable on (1, x).

2. $1 < c < x$ gives $\frac{1}{x} < \frac{1}{c} < 1$.
   Reciprocal reverses inequalities for positive numbers.

3. Multiply by $x - 1 > 0$: $\frac{x-1}{x} < \ln x < x - 1$.
   Direction kept. ∎

HYPOTHESES CHECKLIST (FOR THIS USE OF THE MVT)

- ☐ f continuous on the closed [a, b]
  $\ln$ on $[0, x]$: $\ln 0$ undefined; start at 1, not 0
- ☐ a bound for f' on the whole open interval
  knowing $f'(c)$ for one c is not enough **N12**

YOUR INTUITION LIES · "STRICT OR NOT STRICT?"

The bound on $f'(c)$ is strict ($e^c > 1$ because $c > 0$), so the inequality is strict for $x \neq 0$. At $x = 0$ the interval has length 0, so treat it apart.

WORKED EXAMPLE 1 · FULL (NUMERIC CHECK)

Check the bounds at $x = 2$: $\dfrac{2-1}{2} = 0.5 < \ln 2 = 0.6931 < 1$. At $x = e$: $\dfrac{e-1}{e} = 0.632 < 1 = \ln e < e - 1 = 1.718$. For (a) at $x = 1$: $e = 2.718 > 2$; at $x = -1$: $e^{-1} = 0.368 > 0$.

The MVT point for e^x on $[0, 1]$: $e - 1 = e^c$, so $c = \ln(e-1) \approx 0.5413 \in (0, 1)$.

WORKED EXAMPLE 2 · WITH GAPS

Show $\ln x < x - 1$ for $x > 1$ in two lines. By the MVT on _____, $\ln x = \dfrac{x-1}{c}$ with $1 < c < x$. Since $c > 1$, $\dfrac{1}{c} <$ _____, so $\ln x <$ _____. Answer: $[1, x]$; $\dfrac{1}{c} < 1$; $\ln x < x - 1$.

CHECK YOURSELF

Use part (b) with x replaced by $1 + t$ to show $\dfrac{t}{1+t} < \ln(1+t) < t$ for $t > 0$.

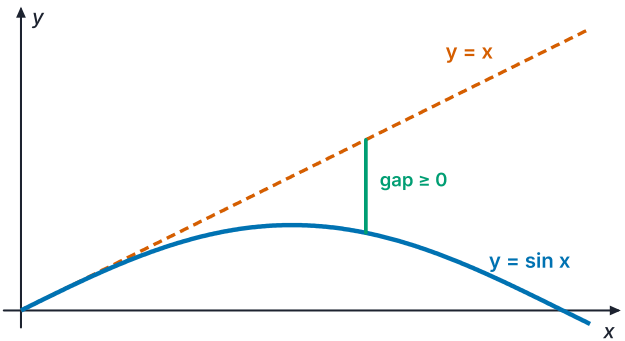
Put $x = 1 + t > 1$ in (b): $\dfrac{x-1}{x} = \dfrac{t}{1+t}$ and $x - 1 = t$. At $t = 1$: $0.5 < 0.693 < 1$. ✓

In the model paper: **Q32** (prove (a), (b)).

T19 Inequalities from the MVT: sine, and Lipschitz

index ↑

$|\sin x - \sin y| \le |x - y|$ and "bounded derivative ⇒ Lipschitz" (MTS-25 Q2, MTS-26 Q3, MTS-22 MCQ 4).



Inequalities from the MVT: sin x ≤ x for x ≥ 0. Apply the MVT to sin on [0, x]: sin x − sin 0 = (cos c)·x ≤ x, since cos c ≤ 1.

RESULTS (BARTLE 6.2.10(B), EX 6.2.6, 6.2.15)

(a) $-x \leq \sin x \leq x$ for all $x \geq 0$; (b) $|\sin x - \sin y| \leq |x - y|$ for all x, y; (c) if f is differentiable on an interval I and $|f'(x)| \leq K$ for all $x \in I$, then $|f(x) - f(u)| \leq K|x - u|$ for all $x, u \in I$: f is **Lipschitz** with constant K.

(c) is the master statement; (a) and (b) are special cases with $f = \sin$, $K = 1$.

SKELETON

Let $x, u \in I$ with $x < u$. f is continuous on $[x, u]$ and differentiable on (x, u). By the MVT, $f(x) - f(u) = f'(c)(x - u)$ for some c between. Take absolute values: $|f(x) - f(u)| = |f'(c)||x - u| \leq$? . ∎

PROOF OF (A), (B), (C)

1. (c) For $x \neq u$ in I, the MVT gives c between x and u with $f(x) - f(u) = f'(c)(x - u)$.
   f is differentiable on all of I, so continuous on $[x, u]$.

2. $|f(x) - f(u)| = |f'(c)|\,|x - u| \leq K|x - u|$.
   $|ab| = |a||b|$, then $|f'(c)| \leq K$ (P09).

3. (b) $f = \sin$, $f' = \cos$, $|\cos| \leq 1$: take $K = 1$.
   Plus Two fact $\sin' = \cos$.

4. (a) Put $y = 0$ in (b): $|\sin x| \leq x$ for $x \geq 0$, i.e. $-x \leq \sin x \leq x$. ∎
   $\sin 0 = 0$ and $|\sin x| \leq x \iff -x \leq \sin x \leq x$.

HYPOTHESES CHECKLIST

- ☐ $|f'| \leq K$ on the whole interval
  $\sqrt{x}$ on $[0, 1]$: $f' = \dfrac{1}{2\sqrt{x}}$ unbounded, not Lipschitz (but uniformly continuous) C4
- ☐ f differentiable *everywhere* on I
  $|x|$ is Lipschitz (K = 1) but not differentiable at 0, so the converse (Lipschitz ⇒ bounded f') is not claimed
- ☐ I an interval
  MVT needs the whole segment $[x, u]$ inside I

YOUR INTUITION LIES · "LIPSCHITZ ⇔ BOUNDED DERIVATIVE"

Only ⇐ is proved. A Lipschitz function can fail to be differentiable ($|x|$). And "bounded derivative" gives uniform continuity *on every interval*, bounded or not: $|\sin x - \sin y| \le |x - y|$ on all of $\mathbb{R}$, so $\sin$ is uniformly continuous on $\mathbb{R}$.

WORKED EXAMPLE 1 · FULL

Check (a) at $x = \pi/2$: $\sin x = 1 < 1.5708 = x$. The MVT point: $\sin\frac{\pi}{2} - 0 = \cos c \cdot \frac{\pi}{2}$ gives $\cos c = \frac{2}{\pi} \approx 0.6366$, so $c = \arccos\frac{2}{\pi} \approx 0.8807 \in (0, \frac{\pi}{2})$.

WORKED EXAMPLE 2 · WITH GAPS (MTS-22 MCQ 4)

Show $f(x) = \frac{1}{1+x^2}$ is Lipschitz on $\mathbb{R}$. $f'(x) =$? . $|f'(x)| = \frac{2|x|}{(1+x^2)^2}$. Its largest value is at $x^2 =$? and equals ? < 1. So $K =$? works and f is uniformly continuous on $\mathbb{R}$.

Answer: $-\frac{2x}{(1+x^2)^2}$; $x^2 = \frac{1}{3}$; $\frac{3\sqrt{3}}{8} \approx 0.6495$; $K = \frac{3\sqrt{3}}{8}$ (or simply $K = 1$).

CHECK YOURSELF

Show $|\arctan x - \arctan y| \le |x - y|$ for all x, y.

$f = \arctan$ has $f'(t) = \frac{1}{1+t^2} \in (0, 1]$, so $|f'| \le 1$. By (c) with $K = 1$.

In the model paper: **Q33** (prove (a), (b), (c)). Link: Module 1, T16 (Lipschitz ⇒ uniformly continuous).

T20 Bernoulli's Inequality

index ↑

> $(1+x)^\alpha \geq 1+\alpha x$ (Bartle 6.2.10(c)).

PUZZLE

Is 1.01^{100} bigger than 2, without a calculator? ($1.01^{100} \geq 1+100(0.01) = 2$.)

BERNOULLI'S INEQUALITY (BARTLE 6.2.10(C))

If $\alpha > 1$, then $(1+x)^\alpha \geq 1+\alpha x$ for all $x > -1$, with equality if and only if $x = 0$.

Picture: the curve $y = (1+x)^\alpha$ lies above its tangent line $y = 1+\alpha x$ at $x = 0$. For $0 < \alpha < 1$ the inequality reverses.

SKELETON

Let $h(t) = (1+t)^\alpha$, so $h'(t) = \alpha(1+t)^{\alpha-1}$ for $t > -1$. **Case** $x > 0$**:** MVT on ? . **Case** $-1 < x < 0$**:** MVT on ? . In each case compare $(1+c)^{\alpha-1}$ with ? . **Case** $x = 0$**:** equality. ∎

PROOF

1. $h(t) = (1+t)^\alpha$, $h'(t) = \alpha(1+t)^{\alpha-1}$ for $t > -1$.
   Known derivative, quoted (Plus Two; Bartle Ex 6.1.10(d), misprinted "(c)" in Ex 6.2.10).

2. $x > 0$. MVT on $[0, x]$: $(1+x)^\alpha - 1 = \alpha(1+c)^{\alpha-1}x$, $0 < c < x$. As $1 + c > 1$ and $\alpha - 1 > 0$, $(1+c)^{\alpha-1} > 1$, so $(1+x)^\alpha - 1 > \alpha x$.
   A number > 1 to a positive power stays > 1; multiply by $\alpha x > 0$.

3. $-1 < x < 0$. MVT on $[x, 0]$: $1 - (1+x)^\alpha = \alpha(1+c)^{\alpha-1}(-x)$, $x < c < 0$. Now $0 < 1 + c < 1$ and $\alpha - 1 > 0$, so $(1+c)^{\alpha-1} < 1$ and $1 - (1+x)^\alpha < -\alpha x$.
   A number in (0, 1) to a positive power is in (0, 1). Here $-x > 0$ keeps the direction.

4. Rearranging gives $(1+x)^\alpha > 1 + \alpha x$ for $x \neq 0$; equality at $x = 0$. ∎
   Both cases are strict.

HYPOTHESES CHECKLIST

☐ $\alpha > 1$
$\alpha = \frac{1}{2}$: $\sqrt{1+x} \leq 1 + \frac{x}{2}$, the reverse (T21)

☐ $x > -1$
$\alpha = \frac{3}{2}$, $x = -2$: $(1+x)^{3/2} = (-1)^{3/2}$ is not a real number

Where each was used. $\alpha - 1 > 0$: makes "> 1 stays > 1, < 1 stays < 1". $x > -1$: keeps $1 + c > 0$ so h is differentiable on the interval.

WORKED EXAMPLE 1 · FULL

$\alpha = \frac{3}{2}$, $x = 1$: $2^{3/2} = 2.828 \geq 1 + 1.5 = 2.5$. $x = -0.5$: $0.5^{3/2} = 0.3536 \geq 1 - 0.75 = 0.25$. Both hold.

WORKED EXAMPLE 2 · WITH GAPS

$\alpha = 3$, $x = -\frac{1}{2}$: left side $(1+x)^3 =$? ; right side $1 + 3x =$? . Inequality holds? ? The proof used the MVT on ? . Answer: $(\frac{1}{2})^3 = \frac{1}{8}$; $1 - \frac{3}{2} = -\frac{1}{2}$; yes, $\frac{1}{8} > -\frac{1}{2}$; on $[x, 0] = [-\frac{1}{2}, 0]$.

CHECK YOURSELF

Use Bernoulli to show $(1.01)^{100} > 2$.

$\alpha = 100 > 1$, $x = 0.01 \neq 0$, so $(1.01)^{100} > 1 + 100(0.01) = 2$ (strict, since $x \neq 0$). True value ≈ 2.7048.

In the model paper: **Q34** (prove Bernoulli, check $\alpha = \frac{3}{2}$).

T21 Weighted AM–GM inequality

index ↑

$a^\alpha b^{1-\alpha} \leq \alpha a + (1-\alpha) b$ (Bartle 6.2.10(d)).

PUZZLE

The average of 2 and 8 is 5. Their geometric mean $\sqrt{2 \cdot 8}$ is 4. The geometric mean is always the smaller. Can calculus prove it?

BARTLE 6.2.10(D)

If $0 < \alpha < 1$ and $x \geq 0$, then $x^\alpha \leq \alpha x + (1 - \alpha)$. Consequently, for $a, b > 0$:

$$a^\alpha b^{1-\alpha} \leq \alpha a + (1 - \alpha) b,$$

with equality if and only if $a = b$.

With $\alpha = \frac{1}{2}$ this is $\sqrt{ab} \leq \frac{a+b}{2}$: geometric mean ≤ arithmetic mean.

SCRATCH · FIND THE FUNCTION AND ITS MINIMUM

Want $x^\alpha \leq \alpha x + 1 - \alpha$, i.e. $\alpha x - x^\alpha \geq \alpha - 1$. Let $g(x) = \alpha x - x^\alpha$ on $[0, \infty)$. Equality seems to happen at $x = 1$: $g(1) = \alpha - 1$. So claim: **g has its minimum at 1**.

$g'(x) = \alpha(1 - x^{\alpha-1})$. Sign: for $0 < x < 1$, $x^{\alpha-1} > 1$ (negative exponent), so $g' < 0$; for $x > 1$, $x^{\alpha-1} < 1$, so $g' > 0$.

PROOF

1. Let $g(x) = \alpha x - x^\alpha$ for $x \geq 0$. g is continuous on $[0, \infty)$ and differentiable on $(0, \infty)$ with $g'(x) = \alpha(1 - x^{\alpha-1})$.
   Power rule for $x > 0$; at 0 only continuity is needed.

2. For $0 < x < 1$: $x^{\alpha-1} > 1$, so $g'(x) < 0$. For $x > 1$: $x^{\alpha-1} < 1$, so $g'(x) > 0$.
   $\alpha - 1 < 0$, so $x \mapsto x^{\alpha-1}$ is decreasing.

3. So g is strictly decreasing on $[0, 1]$ and strictly increasing on $[1, \infty)$: $g(x) \geq g(1) = \alpha - 1$, with equality only at $x = 1$.
   Monotonicity Theorem (T14), strict version; or the First Derivative Test (T15).

4. $\alpha x - x^\alpha \geq \alpha - 1$ gives $x^\alpha \leq \alpha x + (1 - \alpha)$.
   Rearrange.

5. Let $a, b > 0$ and $x = a/b$: $\dfrac{a^\alpha}{b^\alpha} \leq \alpha \dfrac{a}{b} + (1 - \alpha)$. Multiply by $b > 0$: $a^\alpha b^{1-\alpha} \leq \alpha a + (1 - \alpha)b$, with equality iff $a/b = 1$. ∎
   $b \cdot b^{-\alpha} = b^{1-\alpha}$.

HYPOTHESES CHECKLIST

☐ $0 < \alpha < 1$
$\alpha = 2$: $x^2 \leq 2x - 1$ is false ($x^2 - 2x + 1 = (x - 1)^2 \geq 0$ gives the reverse, Bernoulli)

☐ $a, b > 0$
$b = 0$: $x = a/b$ undefined; $a = 0$ gives $0 \leq (1 - \alpha)b$, true but not strict

YOUR INTUITION LIES · "G′ = 0 AT 1, SO DONE"

$g'(1) = 0$ makes 1 a **candidate** only. The sign change of g' (– then +) is what proves a minimum, and monotonicity on each side shows it is the *absolute* minimum N8.

WORKED EXAMPLE 1 · FULL

$\alpha = \frac{1}{3}$, $a = 8$, $b = 1$: $8^{1/3} \cdot 1^{2/3} = 2 \le \frac{1}{3}(8) + \frac{2}{3}(1) = \frac{10}{3} \approx 3.33$. ✓. $\alpha = \frac{1}{2}$, $a = 1$, $b = 4$: $\sqrt{4} = 2 \le \frac{5}{2} = 2.5$ ✓.

WORKED EXAMPLE 2 · WITH GAPS

$\alpha = \frac{2}{3}$, $a = 27$, $b = 8$: $a^{2/3} =$?, $b^{1/3} =$?, product $=$?. Right side $\frac{2}{3} \cdot 27 + \frac{1}{3} \cdot 8 =$?. Check $\le$. Answer: 9; 2; 18; $18 + \frac{8}{3} = \frac{62}{3} \approx 20.67$. $18 \le 20.67$ ✓.

CHECK YOURSELF

Show $\sqrt[3]{x} \le \frac{x}{3} + \frac{2}{3}$ for $x \ge 0$. When is there equality?

Take $\alpha = \frac{1}{3}$ in $x^\alpha \le \alpha x + (1 - \alpha)$. Equality iff $x = 1$. Check $x = 8$: $2 \le \frac{8}{3} + \frac{2}{3} = \frac{10}{3}$ ✓.

In the model paper: Q35 (prove, deduce $\sqrt{ab} \le \frac{a+b}{2}$).

UNIT 2 · BARTLE §6.2 (END)

Derivatives never jump

T22 The local sign lemma

index ↑

The engine inside Darboux's Theorem (Bartle Lemma 6.2.11).

PUZZLE

If the slope at c is positive, are the points just to the right of c *higher* than f(c)? Are they higher than *each other* as you move right? One is true, one is not.

PICTURE IN WORDS

f′(c) > 0 just right of c: values above f(c) ↗ (and just left: below)

f′(c) < 0 just left of c: values above f(c) ↖ (and just right: below)

A positive slope at c means the tangent climbs; the graph follows its tangent closely enough, for a short distance, to stay above (right) or below (left) the height f(c).

LEMMA 6.2.11 · LOCAL SIGN LEMMA

Let I be an interval, $f : I \to \mathbb{R}$, $c \in I$, and suppose f has a derivative at c.

(a) If $f'(c) > 0$, then there is $\delta > 0$ such that $f(x) > f(c)$ for $x \in I$ with $c < x < c + \delta$.

(b) If $f'(c) < 0$, then there is $\delta > 0$ such that $f(x) > f(c)$ for $x \in I$ with $c - \delta < x < c$.

Note (b): the values are **above** f(c) on the **left** (the graph is falling through c). Darboux needs exactly this.

SKELETON · (A)

The quotient $\dfrac{f(x)-f(c)}{x-c}$ tends to $f'(c)>0$, so for $0<|x-c|<\delta$ it is ? . For $c<x<c+\delta$: $f(x)-f(c)=(x-c)\cdot$? >0. ∎

PROOF

1. (a) $\lim\limits_{x\to c}\dfrac{f(x)-f(c)}{x-c}=f'(c)>0$. By the local sign property of limits there is $\delta>0$ such that $\dfrac{f(x)-f(c)}{x-c}>0$ for $x\in I$, $0<|x-c|<\delta$.
   Theorem 4.2.9 (a positive limit forces positive values nearby; Module 1).

2. For $c<x<c+\delta$: $x-c>0$, so $f(x)-f(c)=(x-c)\cdot\dfrac{f(x)-f(c)}{x-c}>0$.
   Positive times positive.

3. (b) $f'(c)<0$: choose δ so the quotient is <0 for $0<|x-c|<\delta$. For $c-\delta<x<c$: $x-c<0$ and quotient <0, so $f(x)-f(c)=(x-c)\cdot(\text{quotient})>0$. ∎
   Negative times negative is positive.

HYPOTHESES CHECKLIST

- ☐ $f'(c) \neq 0$ (strict sign)
  $f(x) = -x^2$ at 0: $f'(0) = 0$ and $f(x) < f(0)$ on both sides
- ☐ $f'(c)$ **exists** (two-sided at an interior point)
  $f(x) = -|x|$ at 0: left quotient is $+1$, right quotient is -1; the right values are *below* f(0), so (a)'s conclusion fails C5
- ☐ (a) says *right* of c, not "everywhere near c"
  left of c the values are lower: $f(x) < f(c)$

Where each was used. Existence of $f'(c)$: the limit whose sign is read. $f'(c) > 0$: starts the sign argument. $x \in I$: the points compared stay in the domain, so it also works at an endpoint of I (that is how Darboux uses it).

YOUR INTUITION LIES · "POSITIVE SLOPE ⇒ INCREASING NEAR C"

The lemma compares $f(x)$ only with the **single value** $f(c)$. It does not say f is increasing on $(c-\delta, c+\delta)$. For $g(x) = x + 2x^2 \sin(1/x)$, $g(0) = 0$, $g'(0) = 1 > 0$ and indeed $g(x) > 0$ just right of 0, but $g'(x) = -1$ at $x = \dfrac{1}{2n\pi}$: g goes down inside every window (T14).

WORKED EXAMPLE 1 · FULL

$f(x) = 3x + x^2$ at $c = 0$. $f'(0) = 3 > 0$. The quotient is $\dfrac{3x + x^2 - 0}{x} = 3 + x$, which is > 0 for $-3 < x < 3$, $x \neq 0$. So $\delta = 3$ works: for $0 < x < 3$, $f(x) = x(3 + x) > 0 = f(0)$. (And for $-3 < x < 0$ the values are below.)

WORKED EXAMPLE 2 · WITH GAPS

$f(x) = x^2 - 2x$ at $c = 0$. $f'(0) =$? < 0. Quotient $= x - 2$, which is < 0 for $x <$? . By (b), for $-\delta < x < 0$ we get $f(x) >$? . Here $\delta =$? . Answer: -2; $x < 2$; $f(x) > f(0) = 0$; $\delta = 2$. Check $x = -1$: $1 + 2 = 3 > 0$ ✓.

CHECK YOURSELF

$f'(c) > 0$. Does the lemma let you conclude $f(x_1) < f(x_2)$ for $c < x_1 < x_2 < c + \delta$? Why or why not?

No. It only compares each x with $f(c)$. Two points both to the right of c are not compared. Counterexample: $x + 2x^2 \sin(1/x)$ at 0 (in the trap box).

In the model paper: **Q20** (state and prove (a)).

T23 Darboux's Theorem

index ↑

A derivative has the intermediate value property, even when it is not continuous.

PUZZLE

You know $f'(0) = 0$ and $f'(2) = 4$. Does f′ have to take the value 1 somewhere between? You are *not* told f′ is continuous. (Yes: this theorem.)

THEOREM 6.2.12 · DARBOUX'S THEOREM

If f is differentiable on $I = [a, b]$ and if k is a number between $f'(a)$ and $f'(b)$, then there is at least one point c in (a, b) such that $f'(c) = k$.

"Between" means either $f'(a) < k < f'(b)$ **or** $f'(a) > k > f'(b)$. Bartle writes out only the first case; the exam wants both.

PROOF PICTURE · IN NUMBERS

$f(x) = x^2$ on $[0, 2]$, $f'(a) = f'(0) = 0$, $f'(b) = f'(2) = 4$, $k = 1$. The auxiliary function $g(x) = kx - f(x) = x - x^2$ has $g'(0) = 1 > 0$ (rising at the left end) and $g'(2) = -3 < 0$ (falling at the right end).

x	0 · 0.5 · 1 · 2
g(x)	0 · 0.25 · 0 · −2

The maximum of g on [0, 2] is at the **interior** point $x = \frac{1}{2}$, where $g' = 0$, i.e. $f'(\frac{1}{2}) = 1 = k$.

SKELETON · CASE $f'(a) < k < f'(b)$

Let $g(x) = kx - f(x)$. g is continuous on [a, b], so it has a maximum at some $c \in [a, b]$ (Max–Min). Since $g'(a) = k - f'(a) > 0$, the maximum is not at ? (Lemma 6.2.11(a)). Since $g'(b) = k - f'(b) < 0$, it is not at ? (Lemma (b)). So c is interior and $g'(c) = 0$, i.e. ? . ∎

SCRATCH · HOW TO FIND G (NEVER COPIED INTO THE EXAM)

Want a point where $f'(c) = k$, i.e. $k - f'(c) = 0$. "derivative equals 0" is what happens at an **interior extremum**. So we need a function whose derivative is $k - f'$: that is $g(x) = kx - f(x)$.

Is there an extremum? g is continuous on [a, b], so yes, a maximum exists. Is it inside? The signs of g' at the ends say the graph climbs away from a and falls into b, so the top is not at the ends.

PROOF (BARTLE, WITH THE MISSING CASE WRITTEN OUT)

1. **Case 1:** $f'(a) < k < f'(b)$. Let $g(x) = kx - f(x)$ on $[a, b]$.
   $g' = k - f'$, so $f'(c) = k$ iff $g'(c) = 0$.

2. g is continuous on [a, b], so g attains a maximum at some $c \in [a, b]$.
   Differentiable ⇒ continuous; Max–Min Theorem 5.3.4.

3. $g'(a) = k - f'(a) > 0$, so by Lemma 6.2.11(a) there are x just to the right of a with $g(x) > g(a)$. Hence the maximum is not at a: $c \neq a$.
   A maximum value cannot be exceeded.

4. $g'(b) = k - f'(b) < 0$, so by Lemma 6.2.11(b) there are x just to the left of b with $g(x) > g(b)$. Hence $c \neq b$.
   Lemma (b): values **above** g(b) on the left. This is why (b) is stated that way.

5. So $c \in (a, b)$ is an interior maximum of g, and $g'(c)$ exists. By the Interior Extremum Theorem, $0 = g'(c) = k - f'(c)$, so $f'(c) = k$.
   Theorem 6.2.1 (T10).

6 **Case 2:** $f'(a) > k > f'(b)$. Put $F = -f$ and $K = -k$. Then $F'(a) = -f'(a) < -k = K$ and $K < -f'(b) = F'(b)$, which is Case 1 for F and K. So there is $c \in (a, b)$ with $F'(c) = K$, i.e. $f'(c) = k$. ∎

Multiplying by −1 reverses the order, turning case 2 into case 1. (Equivalently, maximise $f(x) - kx$.) The case Bartle omits.

HYPOTHESES CHECKLIST

☐ f differentiable at **every** point of [a, b] (one-sided at the ends)

$f(x) = |x|$ on $[-1, 1]$: $f'(-1) = -1 < 0 < 1 = f'(1)$ but $f' = \pm 1$ never 0, since f'(0) does not exist **C5**

☐ the domain is an **interval**

$f(x) = -|x|$ on $[-1, 0) \cup (0, 1]$ (the point 0 removed): $f'(-1) = 1$, $f'(1) = -1$, and $f' = \pm 1$ never 0

☐ f′ **continuous**: *not* needed

$f(x) = x^2 \sin(1/x)$, $f(0) = 0$: f' is discontinuous at 0 yet has the property **C6**

Where each was used. Differentiable on [a, b]: g continuous (Max–Min), and $g'(a)$, $g'(b)$ exist for the lemma. Interval: all of [a, b] is the domain, so the maximum exists. Interior Extremum: converts "max at interior c" into $g'(c) = 0$.

YOUR INTUITION LIES · "NO CONTINUITY OF F′, SO NO IVP"

The Intermediate Value Theorem for *continuous* functions (Module 1) is one reason a function can have the IVP, not the only one. Derivatives can wiggle ($\cos\frac{1}{x}$ terms in f′ of $x^2\sin\frac{1}{x}$ near 0 C6) but cannot jump. Darboux is one-way: it can **rule a function out** as a derivative; it does not by itself prove a function is a derivative.

WORKED EXAMPLE 1 · FULL

$f(x)=x^2$ on $[0,2]$, $k=1$. $f'(0)=0<1<4=f'(2)$. $g(x)=x-x^2$, $g'(x)=1-2x$. $g(0)=0$, $g(\frac{1}{2})=\frac{1}{4}$, $g(2)=-2$. The maximum is at $c=\frac{1}{2}\in(0,2)$, and $f'(\frac{1}{2})=1=k$ ✓.

WORKED EXAMPLE 2 · WITH GAPS

$f(x)=x^3$ on $[0,2]$, $k=3$. $f'(0)=$? , $f'(2)=$? , so k is between. $g(x)=3x-x^3$, $g'=$? $=0$ at $x=$? . Check $g(0), g(1), g(2)$: ? . So $c=$? . Answer: 0; 12; $3-3x^2$; $x=1$; $0, 2, -2$ (max at 1); $c=1$, and $f'(1)=3$ ✓.

CHECK YOURSELF · THE SECOND ORDERING

$f(x) = \sin x$ on $[0, \pi]$, $k = \frac{1}{2}$. Here $f'(0) = 1 > \frac{1}{2} > -1 = f'(\pi)$. Which function do you maximise, and what c do you get?

Case 2: maximise $g(x) = f(x) - kx = \sin x - \frac{x}{2}$ (the Case 1 function for $-f$, $-k$). $g'(x) = \cos x - \frac{1}{2} = 0$ at $x = \frac{\pi}{3}$. $g(0) = 0$, $g(\frac{\pi}{3}) = \frac{\sqrt{3}}{2} - \frac{\pi}{6} \approx 0.342$, $g(\pi) = -\frac{\pi}{2}$: max at $\frac{\pi}{3} \in (0, \pi)$, and $f'(\frac{\pi}{3}) = \cos\frac{\pi}{3} = \frac{1}{2}$ ✓.

In the model paper: **Q39** (state and prove, both orderings).

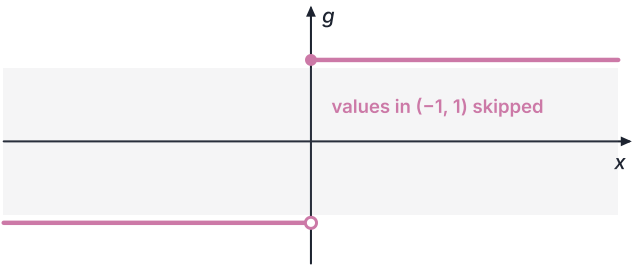
T24 A function that is not a derivative

index ↑

> Jumps are impossible for derivatives (Bartle 6.2.13, Ex 6.2.12).

PUZZLE

The function that is -1 left of 0 and $+1$ right of 0 is perfectly simple. Is there *any* function whose derivative is it? Darboux says no, with no calculation at all.



A jump function (−1 for x < 0, 1 for x ≥ 0) skips all values between −1 and 1. By Darboux it can **not** be the derivative of any function on an interval.

BARTLE 6.2.13 AND EX 6.2.12

The function $g : [-1, 1] \to \mathbb{R}$ defined by $g(x) = -1$ for $-1 \leq x < 0$, $g(0) = 0$, $g(x) = 1$ for $0 < x \leq 1$ is **not** the derivative on $[-1, 1]$ of any function.

The test: a derivative must take every value between any two of its values (Darboux). g takes −1 and 1 but never $\frac{1}{2}$.

SKELETON · "SHOW G IS NOT A DERIVATIVE"

Suppose $g = F'$ on $[a, b]$ for some differentiable F. Pick two values $g(p) < g(q)$ and a number k strictly between them. By Darboux there is c between p and q with $g(c) = k$. But ? . Contradiction. ∎

PROOF

1. Suppose $g = F'$ for a function F differentiable on $[-1, 1]$.
   Contradiction method (M10).

2. $F'(-1) = g(-1) = -1$ and $F'(1) = g(1) = 1$, and $-1 < \frac{1}{2} < 1$.
   Read the values off g.

3. By Darboux there is $c \in (-1, 1)$ with $F'(c) = g(c) = \frac{1}{2}$.
   Theorem 6.2.12 (T23); F is differentiable on [−1, 1].

4. But g takes only the values $-1, 0, 1$. Contradiction. ∎
   $\frac{1}{2}$ is not one of them.

HYPOTHESES CHECKLIST (WHAT THE ARGUMENT NEEDS)

☐ F differentiable on the **whole** interval
$F(x) = |x|$ has $F' = \operatorname{sgn} x$ for $x \neq 0$ but is not differentiable at 0, so sgn is not "F' on [−1, 1]" C9

☐ a value k strictly between two values of g, never taken
$\sin(1/x)$ with value 0 at 0 takes every value between any two of its values: Darboux gives no contradiction C10

YOUR INTUITION LIES · "IT FAILS CONTINUITY, SO IT IS NOT A DERIVATIVE"

Wrong reason. Many derivatives are discontinuous: f′ for $x^2 \sin(1/x)$ C6 . The right test is the **intermediate value property**. And $h(x) = \sin(1/x)$, $h(0) = 0$, is discontinuous at 0 but *does* have the IVP on [0, 1]; Darboux does not rule it out. It is in fact a derivative (it is $F' - 2x \cos \frac{1}{x}$ with $F = x^2 \cos \frac{1}{x}$, and $2x \cos \frac{1}{x}$ is continuous, so has an antiderivative: Module 3) C10 .

SAME DERIVATIVE, DIFFERENT FUNCTIONS (EX 6.2.12)

$h(x) = 0$ for $x < 0$ and $h(x) = 1$ for $x \geq 0$. By Darboux no F has $F' = h$ everywhere (h takes 0 and 1 but not $\frac{1}{2}$). Yet $f_1(x) = \max(x, 0)$ and $f_2(x) = 0$ for $x \leq 0$, $f_2(x) = x + 1$ for $x > 0$ both have derivative $h(x)$ for every $x \neq 0$ (0 on the left, 1 on the right), and $f_2 - f_1 = 0$ for $x \leq 0$, $= 1$ for $x > 0$: **not** a constant. The corollary "equal derivatives ⇒ differ by a constant" needs one *interval* and differentiability at *every* point (T13).

WORKED EXAMPLE 1 · FULL

Heaviside: $H(x) = 0$ $(x < 0)$, $H(x) = 1$ $(x \geq 0)$ on $[-1, 1]$. $H(-1) = 0$, $H(1) = 1$. Take $k = \frac{1}{2}$. H never equals $\frac{1}{2}$ (values 0, 1 only). By Darboux, H is not the derivative of any function on $[-1, 1]$.

WORKED EXAMPLE 2 · WITH GAPS

$g(x) = \lfloor x \rfloor$ (greatest integer) on $[0, 2]$. $g(0) =$? , $g(2) =$? . Take $k =$? . The values of g are only ? , so k is skipped: not a derivative. Answer: 0; 2; $k = \frac{1}{2}$ (or $\frac{3}{2}$); $\{0, 1, 2\}$.

CHECK YOURSELF

f is differentiable on $[0, 1]$ and $f'(x) \neq 0$ for every x. Why must f' have the same sign throughout, and what follows for f?

If $f'(p) < 0 < f'(q)$ for some p, q, Darboux (on [p, q] or [q, p]) gives a point where $f' = 0$: contradiction. So $f' > 0$ everywhere or $f' < 0$ everywhere, and f is strictly monotone (T14).

In the model paper: **Q36** (not a derivative; two functions with the same derivative off 0).

Counterexample library

> Each one breaks a tempting wrong idea. Learn the function, the interval and the reason. (C-codes are the same for the whole course.)

CODE	FUNCTION	WHAT IT SHOWS
C5	$\lvert x \rvert$ at 0	continuous, not differentiable (quotient is ± 1); a minimum with no derivative
C6	$x^2 \sin(1/x)$, value 0 at 0	differentiable everywhere, $f'(0) = 0$, but f' is discontinuous at 0 (values -1 and $+1$ at $\frac{1}{2n\pi}$, $\frac{1}{(2n+1)\pi}$)

C9	sgn, value 0 at 0 (or Heaviside)	jump: skips values, fails the IVP, so **not a derivative**
C10	$\sin(1/x)$, value 0 at 0	has the IVP on [0, 1] yet is discontinuous at 0, and *is* a derivative. (It is **not** an example of "IVP but not a derivative".)
C17	x^3 at 0	$f'(0) = 0$ but no extremum; strictly increasing with a zero derivative
C18	x on [0, 1]; $\lvert x\rvert$ on [−1, 1]; x on [0, 1) with $f(1) = 0$	Rolle/MVT fail: unequal ends; no derivative at 0; no continuity at the right end (in each, $f' \neq 0$ or $\neq$ chord slope)
C19	$x^2 \sin(1/x^2)$, value 0 at 0	differentiable everywhere, f' unbounded near 0 ($f'(x_n) = -2\sqrt{2n\pi}$ at $x_n = 1/\sqrt{2n\pi}$)

Module 2 additions (the N-codes restart in each module):

CODE	FUNCTION	WHAT IT SHOWS
N4	x^2 on $\mathbb{Q}$, 0 off $\mathbb{Q}$	differentiable at 0 only ($\lvert f(x)/x\rvert \leq \lvert x\rvert$); discontinuous elsewhere
N5	0 on (0, 1), 1 on (2, 3); $\dfrac{\lvert x\rvert}{x}$	$f' = 0$ but not constant: the domain is not an interval
N6	$-1/x$ on $\mathbb{R} \setminus \{0\}$	$f' > 0$ but not increasing: $f(-1) = 1 > -1 = f(1)$
N7	x on [0, 1]	extrema at the ends with $f' = 1$: "interior" matters
N8	x^3 and $\lvert x\rvert$ at 0	a critical point is only a candidate
N9	$2x^4 + x^4 \sin(1/x)$	absolute min at 0 but f' changes sign infinitely often near 0: converse of the First Derivative Test is false

N10	$x + 2x^2 \sin(1/x)$	$g'(0) = 1 > 0$ yet g is not monotone near 0 (values -1 and 3 of g')

YOUR INTUITION WILL LIE TO YOU HERE

"If $f'(0) > 0$, then f is increasing near 0." N10 says no. "A derivative must be continuous." C6 says no. "A discontinuous function is not a derivative." C10 says no: the right test is the intermediate value property.

Rule traps without a function: N1 (naive chain rule divides by $f(x) - f(c)$, which can be 0), N3 (piecewise: continuity first, then one-sided derivatives), N11 (quotient needs $g(c) \neq 0$; $(fg)' \neq f'g'$), N12 (MVT gives *some* c; inequalities need a bound over the whole interval), N14 ("increasing" is non-strict).

Which tool?

> Mixed drill. Name the tool first, then check its hypotheses. Answers below.

1. Find $f'(c)$ from the definition.
2. Show f is differentiable at a join of two pieces.
3. Show f is *not* differentiable at c.
4. Differentiate a composite, or $|f|$.
5. Find where f can have a maximum or minimum.
6. Show an equation has exactly one real root.
7. Find the c in "$f(b) - f(a) = f'(c)(b - a)$".
8. Prove $e^x \geq 1 + x$, $\sin x \leq x$, Bernoulli.
9. Show f is increasing / find intervals of increase.
10. Decide whether a critical point is a max, min or neither.
11. Show two functions with equal derivatives differ by a constant.

2. Estimate $\sqrt{105}$ or $\ln 1.1$ without a calculator.
3. Show f is Lipschitz / uniformly continuous from f′.
4. Show f' must take a value between two given values.
5. Show a given function is not a derivative.
6. Prove $a^{\alpha}b^{1-\alpha} \leq \alpha a + (1-\alpha)b$.

ANSWERS

1. Difference quotient; factor $x - c$ (T01, M01, M02). 2. Continuity first, then compare one-sided quotients (T02, M03). 3. Show discontinuity (T02), or two different one-sided quotients, or an unbounded quotient (C5). 4. Chain rule, or $|f|' = \operatorname{sgn}(f)f'$ where $f \neq 0$ (T06, T07). 5. Interior Extremum Theorem: solve $f' = 0$, add points where f′ is undefined, add the ends (T10). 6. Sign change (Location of Roots) plus Rolle for uniqueness (T16). 7. MVT; solve $f'(c) =$ chord slope; reject c outside the interval (T12). 8. MVT on the right interval with a bound for f′ (T18 to T21, M06). 9. Sign of f′ on an interval, M05 (T14). 10. First Derivative Test: sign of f′ on each side (T15). 11. $h = f - g$ has $h' = 0$, so constant (T13). 12. MVT for $\sqrt{}$ or $\ln$ with bounds on c (T17). 13. $|f'| \leq K$ ⇒ Lipschitz ⇒ uniformly continuous (T19; Module 1 T16). 14. Darboux (T23): prove both orderings. 15. A value skipped between two values: Darboux contradiction (T24). 16. $g = \alpha x - x^{\alpha}$ has absolute minimum at 1 (T21).

Theorem sheet

6.1.1 DERIVATIVE

$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$, $c \in I$ (∀ε ∃δ: $0 < |x - c| < \delta$ ⇒ $|\text{quotient} - L| < \varepsilon$).

6.1.2 DIFFERENTIABLE ⇒ CONTINUOUS

$f'(c)$ exists ⇒ f continuous at c. Converse false: $|x|$.

6.1.3 · 6.1.4 RULES

f, g differentiable at c ⇒ αf, $f + g$, fg, and f/g (needs $g(c) \neq 0$): $(fg)' = f'g + fg'$, $(f/g)' = \frac{f'g - fg'}{g^2}$. $Dx^n = nx^{n-1}$.

6.1.5 CARATHÉODORY

f differentiable at c ⇔ ∃ φ on I, continuous at c, $f(x) - f(c) = \varphi(x)(x - c)$. Then $\varphi(c) = f'(c)$.

6.1.6 CHAIN RULE

f differentiable at c, g differentiable at f(c) ⇒ $(g \circ f)'(c) = g'(f(c))f'(c)$.

6.2.1 INTERIOR EXTREMUM · 6.2.2

Relative extremum at an **interior** c and $f'(c)$ exists ⇒ $f'(c) = 0$. If f continuous: $f'(c) = 0$ or does not exist.

6.2.3 ROLLE

f continuous on [a, b], differentiable on (a, b), $f(a) = f(b) = 0 \Rightarrow \exists\, c \in (a, b)$: $f'(c) = 0$.

6.2.4 MVT

f continuous on [a, b], differentiable on (a, b) $\Rightarrow \exists\, c \in (a, b)$: $f(b) - f(a) = f'(c)(b - a)$.

6.2.5 · 6.2.6 · 6.2.7

$f' = 0$ on (a, b), f continuous on [a, b] $\Rightarrow$ constant. $f' = g' \Rightarrow f = g + C$. f differentiable on an interval: increasing $\Leftrightarrow f' \geq 0$; decreasing $\Leftrightarrow f' \leq 0$; $f' > 0 \Rightarrow$ strictly increasing.

6.2.8 FIRST DERIVATIVE TEST

f continuous on [a, b], c interior, differentiable on (a, c) and (c, b). $f' \geq 0$ left, ≤ 0 right $\Rightarrow$ relative max; reversed $\Rightarrow$ relative min.

6.2.9 ROLLE'S CONSEQUENCES (T16, T17)

Between two roots of f lies a root of f'; $f' \neq 0$ on an interval $\Rightarrow$ at most one root. MVT with a bound on f' gives numerical bounds.

6.2.10 INEQUALITIES (T18 TO T21)

$e^x \geq 1 + x$; $\dfrac{x-1}{x} < \ln x < x - 1$ $(x > 1)$; $|\sin x - \sin y| \leq |x - y|$; $|f'| \leq K \Rightarrow |f(x) - f(u)| \leq K|x - u|$; $\alpha > 1$: $(1 + x)^\alpha \geq 1 + \alpha x$ $(x > -1)$; $0 < \alpha < 1$: $a^\alpha b^{1-\alpha} \leq \alpha a + (1 - \alpha)b$.

6.2.11 LOCAL SIGN LEMMA

$f'(c) > 0 \Rightarrow f(x) > f(c)$ for $c < x < c + \delta$.
$f'(c) < 0 \Rightarrow f(x) > f(c)$ for $c - \delta < x < c$.

6.2.12 DARBOUX

f differentiable on [a, b], k between $f'(a)$ and $f'(b)$ (either order) $\Rightarrow \exists\, c \in (a, b)$: $f'(c) = k$.

6.2.13 · EX 6.2.12

A function that skips a value between two of its values (sgn, Heaviside, $\lfloor x \rfloor$) is not a derivative.

BEFORE QUOTING ROLLE OR THE MVT, WRITE THESE TWO LINES

- ☐ f is continuous on the **closed** [a, b]
  (for Rolle also $f(a) = f(b)$)
- ☐ f is differentiable on the **open** (a, b)
  $|x|$ on [−1, 1] and $1 - x^{2/3}$ on [−1, 1] fail here

Attacking a question

3 marks Exact statement or one computation. Always name the theorem and check its hypotheses in one line.

6 marks Statement, Know/Want, skeleton with the key construction filled in (the gap function, $g = kx - f$, the interval for the MVT). 8 to 12 lines.

10 marks Statement word for word; numbered proof with theorem names (Max–Min, Interior Extremum, Rolle, MVT); where each hypothesis was used; counterexample for

a dropped one. If stuck, write the skeleton anyway.

FIVE-MINUTE FINAL CHECK

Did I say "continuous on [a, b], differentiable on (a, b)"? Did I keep the direction of "<" when multiplying by $b - a > 0$? Did I give strict or non-strict correctly? In Darboux, did I treat the second ordering? Is c really inside the open interval?

Past questions → topics

Module 2 had no direct question in the old CBCSS papers (differentiation was taught in Semester 2 calculus). CALC-24 = FYUGP Calculus, MAT1MN102, Nov 2024: related, not this course. MTS-25 = MTS 6B 10, March 2025 · MTS-26 = April 2026 · MTS-22 = March 2022 (and its MCQ paper) · MAT-22 = MAT 6B 09, March 2022 · MAT-21 = March 2021. Others in this module are Bartle's starred exercises.

PAST OR RELATED QUESTION	TOPIC	MODEL Q
CALC-24 Q5 derivative of 3x from the definition	T01	Q2
MAT-22 MCQ 7 "limit" definition vs derivative	T01	Q1
MAT-22 MCQ 4; MAT-21 MCQ 3 limit of $\lvert x\rvert/x$ at 0	T02	Q21
CALC-24 Q13 x^2 / $4\sqrt{x}$ at 1: **not** differentiable (jump 1 vs 4)	T02	Q22
CALC-24 Q10 critical points of $x^3 - 3x + 1$	T10	Q14
CALC-24 Q9 increasing function with example	T14	Q17

CALC-24 Q20(1) intervals of $x^4 - 5x^3 + 9x^2$	T14	Q18
MTS-22 MCQ 5 increases on (0, 1) (adapted; original partly illegible)	T14	Q19
CALC-24 Q18 extrema of $x^2/(x^4 + 16)$	T15	Q15
MAT-21 Q17 $xe^x - 2 = 0$ has a root in (0, 1)	T16	Q30
MTS-25 Q2; MAT-22 Q15 UC but not Lipschitz	T19	Q33
MTS-26 Q3; MAT-21 Q28a define Lipschitz	T19	Q33
MTS-22 MCQ 4 $\frac{1}{1+x^2}$ uniformly continuous	T19	Q33

Every topic → its anchor question

T01 Q1 · T02 Q21 · T03 Q23 · T04 Q7 · T05 Q24 · T06 Q37 · T07 Q10 · T08 Q12 · T09 Q25 · T10 Q26 · T11 Q27 · T12 Q38 · T13 Q28 · T14 Q40 · T15 Q29 · T16 Q30 · T17 Q31 · T18 Q32 · T19 Q33 · T20 Q34 · T21 Q35 · T22 Q20 · T23 Q39 · T24 Q36

Bartle corrections and scope

> Printed errors and gaps in Bartle & Sherbert, and what is outside this module.

MISPRINTS AND GAPS IN THE TEXTBOOK

- **Ex 6.1.7(b)**: the book assumes $f(x) \neq 0$ *and* $f'(x) \neq 0$. Only $f(x) \neq 0$ is needed for $(1/f)' = -f'/f^2$.
- **6.2.7 proof**: the halves are labelled the wrong way. "$f' \geq 0 \Rightarrow$ increasing" is the half that needs the MVT; "increasing $\Rightarrow f' \geq 0$" needs the order theorem for limits.
- **Darboux proof (6.2.12)**: only $f'(a) < k < f'(b)$ is proved. The case $f'(a) > k > f'(b)$ follows by applying the result to $-f$ and $-k$. Write that line in the exam.
- **Ex 6.2.10(c)** refers to Example 6.1.10(c). It should be **6.1.10(d)**, the derivative of x^r.
- **Standard derivatives** ($\sin$, $\cos$, e^x, $\ln x$, x^r) are quoted by Bartle as known and "proved later" (6.1.7(d), Section 8.3). In the exam, quote them as *Plus Two facts*.
- **CALC-24 Q13** (related paper): $f(x) = x^2$ for $x \leq 1$, $4\sqrt{x}$ for $x > 1$ is **not continuous** at 1 (1 against 4), so $f'(1)$ does not exist. The question hides this.

NOT IN THE EXAM

Inverse functions (6.1.8–6.1.10), L'Hospital's rule (6.3) and Taylor's theorem with Lagrange remainder (6.4) are not part of this module. They are only mentioned where an example touches them.